

Snell Envelope Approach and Verification Theorem for Optimal Stopping Problems

and solutions to selected exercises from Stochastic Differential
Equations by Bernt Øksendal at the end

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1 The Time-Homogeneous Case

Problem 1.1 (The optimal stopping problem). *Let X_t be an Itô diffusion on $\mathbb{R}^n$ and let g (the reward function) be a given function on $\mathbb{R}^n$, satisfying*

- a) $g(\xi) \geq 0$ for all $\xi \in \mathbb{R}^n$
- b) g is continuous.

Find a stopping time $\tau^ = \tau^*(x, \omega)$ (called an optimal stopping time) for $\{X_t\}$ such that*

$$\mathbb{E}^x[g(X_{\tau^*})] = \sup_{\tau} \mathbb{E}^x[g(X_{\tau})] \quad \text{for all } x \in \mathbb{R}^n, \quad (2)$$

the sup being taken over all stopping times τ for $\{X_t\}$. I also want to find the corresponding optimal expected reward

$$g^*(x) = \mathbb{E}^x[g(X_{\tau^*})]. \quad (3)$$

Here $g(X_{\tau})$ is to be interpreted as 0 at the points $\omega \in \Omega$ where $\tau(\omega) = \infty$ and as usual $\mathbb{E}^x$ denotes the expectation with respect to the probability law Q^x of the process $X_t; t \geq 0$ starting at $X_0 = x \in \mathbb{R}^n$.

I may regard X_t as the state of a game at time t , each ω corresponds to one sample of the game. For each time t I have the option of stopping the game, thereby obtaining the reward $g(X_t)$, or continue the game in the hope that stopping it at a later time will give a bigger reward. The problem is of course that I do not know what state the game is in at future times, only the probability distribution of the “future”. Mathematically, this means that the possible “stopping” times I consider really are stopping times: The decision whether $\tau \leq t$ or not should only depend on the behaviour of the Brownian motion B_r (driving the process X) up to time t , or perhaps only on the behaviour of X_r up to time t . So, among all possible stopping times τ I am asking for the optimal one, τ^* , which gives the best result “in the long run”, i.e. the biggest expected reward in the sense of (2).

In the following I will outline how a solution to this problem can be obtained using the material from the Dirichlet problem as worked out in Chapter 9 of Øksendal. Later in this chapter I shall see that my discussion of problem (2)–(3) also covers the apparently more general problems

$$g^*(s, x) = \sup_{\tau} \mathbb{E}^{(s, x)}[g(\tau, X_{\tau})] = \mathbb{E}^{(s, x)}[g(\tau^*, X_{\tau^*})] \quad (4)$$

and

$$G^*(s, x) = \sup_{\tau} \mathbb{E}^{(s, x)} \left[\int_0^{\tau} f(t, X_t) dt + g(\tau, X_{\tau}) \right] = \mathbb{E}^{(s, x)} \left[\int_0^{\tau^*} f(t, X_t) dt + g(\tau^*, X_{\tau^*}) \right] \quad (5)$$

where f is a given reward rate function (satisfying certain conditions).

I shall also discuss possible extensions of problem (2)–(3) to cases where g is not necessarily continuous or where g may assume negative values.

A basic concept in the solution of (2)–(3) is the following:

Definition 1.2. A measurable function $f : \mathbb{R}^n \rightarrow [0, \infty]$ is called *supermeanvalued* (w.r.t. X_t) if

$$f(x) \geq \mathbb{E}^x[f(X_\tau)] \quad (6)$$

for all stopping times τ and all $x \in \mathbb{R}^n$.

If, in addition, f is also lower semicontinuous, then f is called *l.s.c. superharmonic* or *just superharmonic* (w.r.t. X_t).

Note that if $f : \mathbb{R}^n \rightarrow [0, \infty]$ is lower semicontinuous then by the Fatou lemma

$$f(x) \leq \mathbb{E}^x \left[\lim_{k \rightarrow \infty} f(X_{\tau_k}) \right] \leq \lim_{k \rightarrow \infty} \mathbb{E}^x[f(X_{\tau_k})], \quad (7)$$

for any sequence $\{\tau_k\}$ of stopping times such that $\tau_k \rightarrow 0$ a.s. P . Combining this with (6) I see that if f is (l.s.c.) superharmonic, then

$$f(x) = \lim_{k \rightarrow \infty} \mathbb{E}^x[f(X_{\tau_k})] \quad \text{for all } x, \quad (8)$$

for all such sequences τ_k .

Remarks. 1) In the literature one often finds a weaker concept of X_t -superharmonicity, defined by the supermeanvalued property (6) plus the stochastic continuity requirement (8). This weaker concept corresponds to the X_t -harmonicity defined in the Dirichlet problem as worked out in Chapter 9 of Øksendal.

2) If $f \in C^2(\mathbb{R}^n)$ it follows from Dynkin's formula that f is superharmonic w.r.t. X_t if and only if

$$\mathcal{A}f \leq 0$$

where $\mathcal{A}$ is the characteristic operator of X_t . This is often a useful criterion.

3) If $X_t = B_t$ is Brownian motion in $\mathbb{R}^n$ then the superharmonic functions for X_t coincide with the (nonnegative) superharmonic functions in classical potential theory.

I state some useful properties of superharmonic and supermeanvalued functions.

Lemma 1.3. a) If f is superharmonic (supermeanvalued) and $\alpha > 0$, then αf is superharmonic (supermeanvalued).

b) If f_1, f_2 are superharmonic (supermeanvalued), then $f_1 + f_2$ is superharmonic (supermeanvalued).

c) If $\{f_j\}_{j \in J}$ is a family of supermeanvalued functions, then $f(x) := \inf_{j \in J} \{f_j(x)\}$ is supermeanvalued if it is measurable (J is any set).

d) If $f_1, f_2, \dots$ are superharmonic (supermeanvalued) functions and $f_k \uparrow f$ pointwise, then f is superharmonic (supermeanvalued).

e) If f is supermeanvalued and $\sigma \leq \tau$ are stopping times, then $\mathbb{E}^x[f(X_\sigma)] \geq \mathbb{E}^x[f(X_\tau)]$.

f) If f is supermeanvalued and H is a Borel set, then $\tilde{f}(x) := \mathbb{E}^x[f(X_{\tau_H})]$ is supermeanvalued.

Proof. a) and b) are straightforward.

c) Suppose f_j is supermeanvalued for all $j \in J$. Then

$$f_j(x) \geq \mathbb{E}^x[f_j(X_\tau)] \geq \mathbb{E}^x[f(X_\tau)] \quad \text{for all } j.$$

So $f(x) = \inf f_j(x) \geq \mathbb{E}^x[f(X_\tau)]$, as required.

d) Suppose f_j is supermeanvalued, $f_j \uparrow f$. Then

$$f(x) \geq f_j(x) \geq \mathbb{E}^x[f_j(X_\tau)] \quad \text{for all } j, \text{ so}$$

$$f(x) \geq \lim_{j \rightarrow \infty} \mathbb{E}^x[f_j(X_\tau)] = \mathbb{E}^x[f(X_\tau)],$$

by monotone convergence. Hence f is supermeanvalued. If each f_j is also lower semicontinuous then if $y_k \rightarrow x$ as $k \rightarrow \infty$ I have

$$f_j(x) \leq \lim_{k \rightarrow \infty} f_j(y_k) \leq \lim_{k \rightarrow \infty} f(y_k) \quad \text{for each } j.$$

Hence, by letting $j \rightarrow \infty$,

$$f(x) \leq \lim_{k \rightarrow \infty} f(y_k).$$

e) If f is supermeanvalued I have by the Markov property when $t > s$

$$\mathbb{E}^x[f(X_t)|\mathcal{F}_s] = \mathbb{E}^{X_s}[f(X_{t-s})] \leq f(X_s), \quad (9)$$

i.e. the process

$$\zeta_t = f(X_t)$$

is a supermartingale w.r.t. the σ -algebras $\mathcal{F}_t$ generated by $\{B_r; r \leq t\}$. Therefore, by Doob's optional sampling theorem I have

$$\mathbb{E}^x[f(X_\sigma)] \geq \mathbb{E}^x[f(X_\tau)]$$

for all stopping times σ, τ with $\sigma \leq \tau$ a.s. Q^x .

f) Suppose f is supermeanvalued. By the strong Markov property I have, for any stopping time α ,

$$\begin{aligned} \mathbb{E}^x[\tilde{f}(X_\alpha)] &= \mathbb{E}^x[\mathbb{E}^{X_\alpha}[f(X_{\tau_H})]] = \mathbb{E}^x[\mathbb{E}^x[\theta_\alpha f(X_{\tau_H})|\mathcal{F}_\alpha]] \\ &= \mathbb{E}^x[\theta_\alpha f(X_{\tau_H})] = \mathbb{E}^x[f(X_{\tau_H^\alpha})] \end{aligned} \quad (10)$$

where $\tau_H^\alpha = \inf\{t > \alpha; X_t \notin H\}$. Since $\tau_H^\alpha \geq \tau_H$ I have by e)

$$\mathbb{E}^x[\tilde{f}(X_\alpha)] \leq \mathbb{E}^x[f(X_{\tau_H})] = \tilde{f}(x),$$

so $\tilde{f}$ is supermeanvalued. □

The following concepts are fundamental:

Definition 1.4. Let h be a real measurable function on $\mathbb{R}^n$. If f is a superharmonic (supermeanvalued) function and $f \geq h$ I say that f is a superharmonic (supermeanvalued) majorant of h (w.r.t. X_t). The function

$$\hat{h}(x) = \inf_f \{f(x); x \in \mathbb{R}^n\}, \quad (11)$$

the inf being taken over all supermeanvalued majorants f of h , is called the least supermeanvalued majorant of h .

Similarly, suppose there exists a function $\hat{h}$ such that (i) $\hat{h}$ is a superharmonic majorant of h and (ii) if f is any other superharmonic majorant of h then $\hat{h} \leq f$. Then $\hat{h}$ is called the least superharmonic majorant of h .

Note that by Lemma 1.3 c) the function $\hat{h}$ is supermeanvalued if it is measurable. Moreover, if h is lower semicontinuous, then $\hat{h}$ exists and $\hat{h} = h$. Later I will prove that if g is nonnegative (or lower bounded) and lower semicontinuous, then $\hat{g}$ exists and $\hat{g} = g$ (Theorem 1.7).

Let $g \geq 0$ and let f be a supermeanvalued majorant of g . Then if τ is a stopping time

$$f(x) \geq \mathbb{E}^x[f(X_\tau)] \geq \mathbb{E}^x[g(X_\tau)].$$

So

$$f(x) \geq \sup_{\tau} \mathbb{E}^x[g(X_\tau)] = g^*(x).$$

Therefore I always have

$$\hat{g}(x) \geq g^*(x) \quad \text{for all } x \in \mathbb{R}^n. \quad (12)$$

What is not so easy to see is that the converse inequality also holds, i.e. that in fact

$$\hat{g} = g^*. \quad (13)$$

I will prove this after I have established a useful iterative procedure for calculating $\hat{g}$. Before I give such a procedure let me introduce a concept which is related to superharmonic functions:

Definition 1.5. A lower semicontinuous function $f : \mathbb{R}^n \rightarrow [0, \infty]$ is called excessive (w.r.t. X_t) if

$$f(x) \geq \mathbb{E}^x[f(X_s)] \quad \text{for all } s \geq 0, x \in \mathbb{R}^n. \quad (14)$$

It is clear that a superharmonic function must be excessive. What is not so obvious, is that the converse also holds:

Theorem 1.6. Let $f : \mathbb{R}^n \rightarrow [0, \infty]$. Then f is excessive w.r.t. X_t if and only if f is superharmonic w.r.t. X_t .

Proof in a special case. Let L be the differential operator associated to X , so that L coincides with the generator $\mathcal{A}$ of X on C_0^2 . I only prove the theorem

in the special case when $f \in C^2(\mathbb{R}^n)$ and Lf is bounded: Then by Dynkin's formula I have

$$\mathbb{E}^x[f(X_t)] = f(x) + \mathbb{E}^x \left[\int_0^t Lf(X_r) dr \right] \quad \text{for all } t \geq 0,$$

so if f is excessive then $Lf \leq 0$. Therefore, if τ is a stopping time I get

$$\mathbb{E}^x[f(X_{t \wedge \tau})] \leq f(x) \quad \text{for all } t \geq 0.$$

Letting $t \rightarrow \infty$ I see that f is superharmonic. $\square$

The first iterative procedure for the least superharmonic majorant $\hat{g}$ of g is the following:

Theorem 1.7 (Construction of the least superharmonic majorant). *Let $g = g_0$ be a nonnegative, lower semicontinuous function on $\mathbb{R}^n$ and define inductively*

$$g_n(x) = \sup_{t \in S_n} \mathbb{E}^x[g_{n-1}(X_t)], \quad (15)$$

where $S_n = \{k \cdot 2^{-n}; 0 \leq k \leq 4^n\}$, $n = 1, 2, \dots$. Then $g_n \uparrow \hat{g}$ and $\hat{g}$ is the least superharmonic majorant of g . Moreover, $\hat{g} = g$.

Proof. Note that $\{g_n\}$ is increasing. Define $\check{g}(x) = \lim_{n \rightarrow \infty} g_n(x)$. Then

$$\check{g}(x) \geq g_n(x) \geq \mathbb{E}^x[g_{n-1}(X_t)] \quad \text{for all } n \text{ and all } t \in S_n.$$

Hence

$$\check{g}(x) \geq \lim_{n \rightarrow \infty} \mathbb{E}^x[g_{n-1}(X_t)] = \mathbb{E}^x[\check{g}(X_t)] \quad (16)$$

for all $t \in S = \bigcup_{n=1}^{\infty} S_n$. Since $\check{g}$ is an increasing limit of lower semicontinuous functions, $\check{g}$ is lower semicontinuous. Fix $t \in \mathbb{R}$ and choose $t_k \in S$ such that $t_k \rightarrow t$. Then by (16), the Fatou lemma and lower semicontinuity

$$\check{g}(x) \geq \lim_{k \rightarrow \infty} \mathbb{E}^x[\check{g}(X_{t_k})] \geq \mathbb{E}^x \left[\lim_{k \rightarrow \infty} \check{g}(X_{t_k}) \right] \geq \mathbb{E}^x[\check{g}(X_t)].$$

So $\check{g}$ is an excessive function. Therefore $\check{g}$ is superharmonic by Theorem 1.6 and hence $\check{g}$ is a superharmonic majorant of g . On the other hand, if f is any supermeanvalued majorant of g , then clearly by induction

$$f(x) \geq g_n(x) \quad \text{for all } n$$

and so $f(x) \geq \check{g}(x)$. This proves that $\check{g}$ is the least supermeanvalued majorant of g . So $\check{g} = \hat{g}$. $\square$

It is a consequence of Theorem 1.7 that I may replace the finite sets S_n by the whole interval $[0, \infty]$:

Corollary 1.8. Define $h_0 = g$ and inductively

$$h_n(x) = \sup_{t \geq 0} \mathbb{E}^x[h_{n-1}(X_t)]; \quad n = 1, 2, \dots$$

Then $h_n \uparrow \hat{g}$.

Proof. Let $h = \lim_{n \rightarrow \infty} h_n$. Then clearly $h \geq \check{g} = \hat{g}$. On the other hand, since $\hat{g}$ is excessive I have

$$\hat{g}(x) \geq \sup_{t \geq 0} \mathbb{E}^x[\hat{g}(X_t)].$$

So by induction

$$\hat{g} \geq h_n \quad \text{for all } n.$$

Thus $\hat{g} = h$ and the proof is complete. $\square$

The following theorem tells us that the Snell envelope, used in stochastics and mathematical finance, is the smallest supermartingale dominating a stochastic process. The Snell envelope is named after James Laurie Snell.

Theorem 1.9 (Existence theorem for optimal stopping). *Let g^* denote the optimal reward and $\hat{g}$ the least superharmonic majorant of a continuous reward function $g \geq 0$.*

a) Then

$$g^*(x) = \hat{g}(x). \quad (17)$$

b) For $\epsilon > 0$ let

$$D_\epsilon = \{x; g(x) < \hat{g}(x) - \epsilon\}. \quad (18)$$

Suppose g is bounded. Then stopping at the first time τ_ϵ of exit from D_ϵ is close to being optimal, in the sense that

$$|g^*(x) - \mathbb{E}^x[g(X_{\tau_\epsilon})]| \leq 2\epsilon \quad (19)$$

for all x .

c) For arbitrary continuous $g \geq 0$ let

$$D = \{x; g(x) < g^*(x)\} \quad (\text{the continuation region}). \quad (20)$$

For $N = 1, 2, \dots$ define $g_N = g \wedge N$, $D_N = \{x; g_N(x) < \hat{g}_N(x)\}$ and $\sigma_N = \tau_{D_N}$. Then $D_N \subset D_{N+1}$, $D_N \subset D \cap g^{-1}([0, N))$, $D = \bigcup_N D_N$. If $\sigma_N < \infty$ a.s. Q^x for all N then

$$g^*(x) = \lim_{N \rightarrow \infty} \mathbb{E}^x[g(X_{\sigma_N})]. \quad (21)$$

d) In particular, if $\tau_D < \infty$ a.s. Q^x and the family $\{g(X_{\sigma_N})\}_N$ is uniformly integrable w.r.t. Q^x , then

$$g^*(x) = \mathbb{E}^x[g(X_{\tau_D})]$$

and $\tau^* = \tau_D$ is an optimal stopping time.

Proof. First assume that g is bounded and define

$$\tilde{g}_\epsilon(x) = \mathbb{E}^x[\hat{g}(X_{\tau_\epsilon})] \quad \text{for } \epsilon > 0. \quad (22)$$

Then $\tilde{g}_\epsilon$ is supermeanvalued by Lemma 1.3 f). I claim that

$$g(x) \leq \tilde{g}_\epsilon(x) + \epsilon \quad \text{for all } x. \quad (23)$$

To see this suppose

$$\beta := \sup_x \{g(x) - \tilde{g}_\epsilon(x)\} > \epsilon. \quad (24)$$

Then for all $\eta > 0$ I can find x_0 such that

$$g(x_0) - \tilde{g}_\epsilon(x_0) \geq \beta - \eta. \quad (25)$$

On the other hand, since $\tilde{g}_\epsilon + \beta$ is a supermeanvalued majorant of g , I have

$$\hat{g}(x_0) \leq \tilde{g}_\epsilon(x_0) + \beta. \quad (26)$$

Combining (25) and (26) I get

$$\hat{g}(x_0) \leq g(x_0) + \eta. \quad (27)$$

Consider the two possible cases:

Case 1: $\tau_\epsilon > 0$ a.s. Q^{x_0} . Then by (27) and the definition of D_ϵ

$$g(x_0) + \eta \geq \hat{g}(x_0) \geq \mathbb{E}^{x_0}[\hat{g}(X_{t \wedge \tau_\epsilon})] \geq \mathbb{E}^{x_0}[(g(X_t) + \epsilon)\chi_{\{t < \tau_\epsilon\}}] \quad \text{for all } t > 0.$$

Hence by the Fatou lemma and lower semicontinuity of g

$$\begin{aligned} g(x_0) + \eta &\geq \lim_{t \rightarrow 0} \mathbb{E}^{x_0}[(g(X_t) + \epsilon)\chi_{\{t < \tau_\epsilon\}}] \\ &\geq \mathbb{E}^{x_0} \left[\lim_{t \rightarrow 0} (g(X_t) + \epsilon)\chi_{\{t < \tau_\epsilon\}} \right] \geq g(x_0) + \epsilon. \end{aligned}$$

This is a contradiction if $\eta < \epsilon$.

Case 2: $\tau_\epsilon = 0$ a.s. Q^{x_0} . Then $\tilde{g}_\epsilon(x_0) = \hat{g}(x_0)$, so $g(x_0) \leq \tilde{g}_\epsilon(x_0)$, contradicting (25) for $\eta < \beta$.

Therefore (24) leads to a contradiction. Thus (23) is proved and I conclude that $\tilde{g}_\epsilon + \epsilon$ is a supermeanvalued majorant of g . Therefore

$$\hat{g} \leq \tilde{g}_\epsilon + \epsilon = \mathbb{E}[\hat{g}(X_{\tau_\epsilon})] + \epsilon \leq \mathbb{E}[(g + \epsilon)(X_{\tau_\epsilon})] + \epsilon \leq g^* + 2\epsilon \quad (28)$$

and since ϵ was arbitrary I have by (12)

$$\hat{g} = g^*.$$

If g is not bounded, let

$$g_N = \min(N, g), \quad N = 1, 2, \dots$$

and as before let $\hat{g}_N$ be the least superharmonic majorant of g_N . Then

$$g^* \geq g_N^* = \hat{g}_N \uparrow h \text{ as } N \rightarrow \infty, \text{ where } h \geq \hat{g}$$

since h is a superharmonic majorant of g . Thus $h = \hat{g} = g^*$ and this proves (17) for general g . From (28) and (17) I obtain (19).

Finally, to obtain c) and d) let me again first assume that g is bounded. Then, since

$$\tau_\epsilon \uparrow \tau_D \text{ as } \epsilon \downarrow 0$$

and $\tau_D < \infty$ a.s I have

$$\mathbb{E}^x[g(X_{\tau_\epsilon})] \rightarrow \mathbb{E}^x[g(X_{\tau_D})] \text{ as } \epsilon \downarrow 0, \quad (29)$$

and hence by (28) and (17)

$$g^*(x) = \mathbb{E}^x[g(X_{\tau_D})] \text{ if } g \text{ is bounded.} \quad (30)$$

Finally, if g is not bounded define

$$h = \lim_{N \rightarrow \infty} \hat{g}_N.$$

Then h is superharmonic by Lemma 1.3 d) and since $\hat{g}_N \leq \hat{g}$ for all N I have $h \leq \hat{g}$. On the other hand $g_N \leq \hat{g}_N \leq h$ for all N and therefore $g \leq h$. Since $\hat{g}$ is the least superharmonic majorant of g I conclude that

$$h = \hat{g}. \quad (31)$$

Hence by (30), (31) I obtain (21):

$$g^*(x) = \lim_{N \rightarrow \infty} \hat{g}_N(x) = \lim_{N \rightarrow \infty} \mathbb{E}^x[g_N(X_{\sigma_N})] \leq \lim_{N \rightarrow \infty} \mathbb{E}^x[g(X_{\sigma_N})] \leq g^*(x).$$

Note that $\hat{g}_N \leq N$ everywhere, so if $g_N(x) < \hat{g}_N(x)$ then $g_N(x) < N$ and therefore $g(x) = g_N(x) < \hat{g}_N(x) \leq \hat{g}(x)$ and $g_{N+1}(x) = g_N(x) < \hat{g}_N(x) \leq \hat{g}_{N+1}(x)$. Hence $D_N \subset D \cap \{x; g(x) < N\}$ and $D_N \subset D_{N+1}$ for all N . So by (31) I conclude that D is the increasing union of the sets D_N ; $N = 1, 2, \dots$. Therefore

$$\tau_D = \lim_{N \rightarrow \infty} \sigma_N.$$

So by (21) and uniform integrability I have

$$\begin{aligned} \hat{g}(x) &= \lim_{N \rightarrow \infty} \hat{g}_N(x) = \lim_{N \rightarrow \infty} \mathbb{E}^x[g_N(X_{\sigma_N})] \\ &= \mathbb{E}^x \left[\lim_{N \rightarrow \infty} g_N(X_{\sigma_N}) \right] = \mathbb{E}^x[g(X_{\tau_D})], \end{aligned}$$

and the proof of Theorem 1.9 is complete. $\square$

Remarks.

- 1) Note that the sets D , D_ϵ and D_N are open, since $\hat{g} = g^*$ is lower semicontinuous and g is continuous.
- 2) By inspecting the proof of a) I see that (17) holds under the weaker assumption that $g \geq 0$ is lower semicontinuous.

The following consequence of Theorem 1.9 is often useful:

Corollary 1.10. *Suppose there exists a Borel set H such that*

$$\tilde{g}_H(x) := \mathbb{E}^x[g(X_{\tau_H})]$$

is a supermeanvalued majorant of g . Then

$$g^*(x) = \tilde{g}_H(x), \quad \text{so } \tau^* = \tau_H \text{ is optimal.}$$

Proof. If $\tilde{g}_H$ is a supermeanvalued majorant of g then clearly

$$g(x) \leq \tilde{g}_H(x).$$

On the other hand I of course have

$$\tilde{g}_H(x) \leq \sup_{\tau} \mathbb{E}^x[g(X_{\tau})] = g^*(x),$$

so $g^* = \tilde{g}_H$ by Theorem 1.7 and Theorem 1.9 a). □

Corollary 1.11. *Let*

$$D = \{x; g(x) < \hat{g}(x)\}$$

and put

$$\tilde{g}(x) = \tilde{g}_D(x) = \mathbb{E}^x[g(X_{\tau_D})].$$

If $\tilde{g} \geq g$ then $\tilde{g} = g^$.*

Proof. Since $X_{\tau_D} \notin D$ I have $g(X_{\tau_D}) \geq \hat{g}(X_{\tau_D})$ and therefore $g(X_{\tau_D}) = \hat{g}(X_{\tau_D})$, a.s. Q^x . So $\tilde{g}(x) = \mathbb{E}^x[\hat{g}(X_{\tau_D})]$ is supermeanvalued since $\hat{g}$ is, and the result follows from Corollary 1.10. □

Theorem 1.9 gives a sufficient condition for the existence of an optimal stopping time τ^* . Unfortunately, τ^* need not exist in general. For example, if

$$X_t = t \quad \text{for } t \geq 0 \quad (\text{deterministic})$$

and

$$g(\xi) = \frac{\xi^2}{1 + \xi^2}; \quad \xi \in \mathbb{R}$$

then $g^*(x) = 1$, but there is no stopping time τ such that

$$\mathbb{E}^x[g(X_{\tau})] = 1.$$

However, I can prove that if an optimal stopping time τ^* exists, then the stopping time given in Theorem 1.9 is optimal:

Theorem 1.12 (Uniqueness theorem for optimal stopping). *Define as before*

$$D = \{x; g(x) < g^*(x)\} \subset \mathbb{R}^n.$$

Suppose there exists an optimal stopping time $\tau^ = \tau^*(x, \omega)$ for the problem (2) for all x . Then*

$$\tau^* \geq \tau_D \quad \text{for all } x \in D \quad (32)$$

and

$$g^*(x) = \mathbb{E}^x[g(X_{\tau_D})] \quad \text{for all } x \in \mathbb{R}^n. \quad (33)$$

Hence τ_D is an optimal stopping time for the problem (2).

Proof. Choose $x \in D$. Let τ be an $\mathcal{F}_t$ -stopping time and assume $Q^x[\tau < \tau_D] > 0$. Since $g(X_\tau) < g^*(X_\tau)$ if $\tau < \tau_D$ and $g \leq g^*$ always, I have

$$\begin{aligned} \mathbb{E}^x[g(X_\tau)] &= \int_{\tau < \tau_D} g(X_\tau) dQ^x + \int_{\tau \geq \tau_D} g(X_\tau) dQ^x \\ &< \int_{\tau < \tau_D} g^*(X_\tau) dQ^x + \int_{\tau \geq \tau_D} g^*(X_\tau) dQ^x = \mathbb{E}^x[g^*(X_\tau)] \leq g^*(x), \end{aligned}$$

since g^* is superharmonic. This proves (32).

To obtain (33) I first choose $x \in D$. Since $\hat{g}$ is superharmonic I have by (32) and Lemma 1.3 e)

$$\begin{aligned} g^*(x) &= \mathbb{E}^x[g(X_{\tau^*})] \leq \mathbb{E}^x[\hat{g}(X_{\tau^*})] \leq \mathbb{E}^x[\hat{g}(X_{\tau_D})] \\ &= \mathbb{E}^x[g(X_{\tau_D})] \leq g^*(x), \quad \text{which proves (33) for } x \in D. \end{aligned}$$

Next, choose $x \in \partial D$ to be an irregular boundary point of D . Then $\tau_D > 0$ a.s. Q^x . Let $\{\alpha_k\}$ be a sequence of stopping times such that $0 < \alpha_k < \tau_D$ and $\alpha_k \rightarrow 0$ a.s. Q^x , as $k \rightarrow \infty$. Then $X_{\alpha_k} \in D$ so by (32) and the strong Markov property

$$\mathbb{E}^x[g(X_{\tau_D})] = \mathbb{E}^x[\theta_{\alpha_k} g(X_{\tau_D})] = \mathbb{E}^x[\mathbb{E}^{X_{\alpha_k}}[g(X_{\tau_D})]] = \mathbb{E}^x[g^*(X_{\alpha_k})] \quad \text{for all } k.$$

Hence by lower semicontinuity and the Fatou lemma

$$g^*(x) \leq \mathbb{E}^x \left[\lim_{k \rightarrow \infty} g^*(X_{\alpha_k}) \right] \leq \lim_{k \rightarrow \infty} \mathbb{E}^x[g^*(X_{\alpha_k})] = \mathbb{E}^x[g(X_{\tau_D})].$$

Finally, if $x \in \partial D$ is a regular boundary point of D or if $x \notin D$ I have $\tau_D = 0$ a.s. Q^x and hence $g^*(x) = \mathbb{E}^x[g(X_{\tau_D})]$. $\square$

Remark. The following observation is sometimes useful:
Let $\mathcal{A}$ be the characteristic operator of X . Assume $g \in C^2(\mathbb{R}^n)$. Define

$$U = \{x; \mathcal{A}g(x) > 0\}. \quad (34)$$

Then, with D as before, (20),

$$U \subset D. \quad (35)$$

Consequently, from (32) I conclude that it is never optimal to stop the process before it exits from U . But there may be cases when $U \neq D$, so that it is optimal to proceed beyond U before stopping. (This is in fact the typical situation.) See e.g. Example 2.2.

To prove (35) choose $x \in U$ and let τ_0 be the first exit time from a bounded open set $W \ni x, W \subset U$. Then by Dynkin's formula, for $u > 0$

$$\mathbb{E}^x[g(X_{\tau_0 \wedge u})] = g(x) + \mathbb{E}^x \left[\int_0^{\tau_0 \wedge u} \mathcal{A}g(X_s) ds \right] > g(x)$$

so $g(x) < g^*(x)$ and therefore $x \in D$.

Example 1.13. Let $X_t = B_t$ be a Brownian motion in $\mathbb{R}^2$. Using that B_t is recurrent in $\mathbb{R}^2$ one can show that the only (nonnegative) superharmonic functions in $\mathbb{R}^2$ are the constants. Therefore

$$g^*(x) = \|g\|_\infty := \sup\{g(y); y \in \mathbb{R}^2\} \quad \text{for all } x.$$

So if g is unbounded then $g^* = \infty$ and no optimal stopping time exists. Assume therefore that g is bounded. The continuation region is

$$D = \{x; g(x) < \|g\|_\infty\},$$

so if ∂D is a polar set i.e. $\text{cap}(\partial D) = 0$, where cap denotes the logarithmic capacity, then $\tau_D = \infty$ a.s. and no optimal stopping exists. On the other hand, if $\text{cap}(\partial D) > 0$ then $\tau_D < \infty$ a.s. and

$$\mathbb{E}^x[g(B_{\tau_D})] = \|g\|_\infty = g^*(x),$$

so $\tau^* = \tau_D$ is optimal.

Example 1.14. The situation is different in $\mathbb{R}^n$ for $n \geq 3$.

a) To illustrate this let $X_t = B_t$ be Brownian motion in $\mathbb{R}^3$ and let the reward function be

$$g(\xi) = \begin{cases} |\xi|^{-1} & \text{for } |\xi| \geq 1 \\ 1 & \text{for } |\xi| < 1 \end{cases}; \quad \xi \in \mathbb{R}^3.$$

Then g is superharmonic (in the classical sense) in $\mathbb{R}^3$, so $g^* = g$ everywhere and the best policy is to stop immediately, no matter where the starting point is.

b) Let me change g to

$$h(x) = \begin{cases} |x|^{-\alpha} & \text{for } |x| \geq 1 \\ 1 & \text{for } |x| < 1 \end{cases}$$

for some $\alpha > 1$. Let $H = \{x; |x| > 1\}$ and define

$$\tilde{h}(x) = \mathbb{E}^x[h(B_{\tau_H})] = P^x[\tau_H < \infty].$$

Then

$$\tilde{h}(x) = \begin{cases} 1 & \text{if } |x| \leq 1 \\ |x|^{-1} & \text{if } |x| > 1 \end{cases},$$

i.e. $\tilde{h} = g$ (defined in a)), which is a superharmonic majorant of h . Therefore by Corollary 1.10

$$h^* = \tilde{h} = g,$$

$H = D$ and $\tau^* = \tau_H$ is an optimal stopping time.

Reward Functions Assuming Negative Values

The results I have obtained so far regarding the problem (2)–(3) are based on the assumptions (1). To some extent these assumptions can be relaxed, although neither can be removed completely. For example, I have noted that Theorem 1.9 a) still holds if $g \geq 0$ is only assumed to be lower semicontinuous.

The nonnegativity assumption on g can also be relaxed. First of all, note that if g is bounded below, say $g \geq -M$ where $M > 0$ is a constant, then I can put

$$g_1 = g + M \geq 0$$

and apply the theory to g_1 . Since

$$\mathbb{E}^x[g(X_\tau)] = \mathbb{E}^x[g_1(X_\tau)] - M \quad \text{if } \tau < \infty \text{ a.s.,}$$

I have $g^*(x) = g_1^*(x) - M$, so the problem can be reduced to the optimal stopping problem for the nonnegative function g_1 .

If g is not bounded below, then problem (2)–(3) is not well-defined unless

$$\mathbb{E}^x[g^-(X_\tau)] < \infty \quad \text{for all } \tau \tag{36}$$

where

$$g^-(x) = -\min(g(x), 0).$$

If I assume that g satisfies the stronger condition that

$$\text{the family } \{g^-(X_\tau); \tau \text{ stopping time}\} \text{ is uniformly integrable} \tag{37}$$

then basically all the theory from the nonnegative case carries over. See also Theorem 4.1.

2 The Time-Inhomogeneous Case

Let me now consider the case when the reward function g depends on both time and space, i.e.

$$g = g(t, x) : \mathbb{R} \times \mathbb{R}^n \rightarrow [0, \infty), \quad g \text{ is continuous.} \tag{38}$$

Then the problem is to find $g_0(x)$ and τ^* such that

$$g_0(x) = \sup_{\tau} \mathbb{E}^x[g(\tau, X_{\tau})] = \mathbb{E}^x[g(\tau^*, X_{\tau^*})]. \quad (39)$$

To reduce this case to the original case (2)–(3) I proceed as follows: Suppose the Itô diffusion $X_t = X_t^x$ has the form

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t; \quad t \geq 0, \quad X_0 = x$$

where $b : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ are given functions satisfying standard existence and uniqueness conditions and B_t is m -dimensional Brownian motion. Define the Itô diffusion $Y_t = Y_t^{(s,x)}$ in $\mathbb{R}^{n+1}$ by

$$Y_t = \begin{bmatrix} s+t \\ X_t^x \end{bmatrix}; \quad t \geq 0. \quad (40)$$

Then

$$dY_t = \begin{bmatrix} 1 \\ b(X_t) \end{bmatrix} dt + \begin{bmatrix} 0 \\ \sigma(X_t) \end{bmatrix} dB_t = \hat{b}(Y_t)dt + \hat{\sigma}(Y_t)dB_t \quad (41)$$

where

$$\hat{b}(\eta) = \hat{b}(t, \xi) = \begin{bmatrix} 1 \\ b(\xi) \end{bmatrix} \in \mathbb{R}^{n+1}, \quad \hat{\sigma}(\eta) = \hat{\sigma}(t, \xi) = \begin{bmatrix} 0 & \cdots & 0 \\ & \sigma(\xi) & \end{bmatrix} \in \mathbb{R}^{(n+1) \times m},$$

with $\eta = (t, \xi) \in \mathbb{R} \times \mathbb{R}^n$.

So Y_t is an Itô diffusion starting at $y = (s, x)$. Let $R_y = R_{(s,x)}$ denote the probability law of $\{Y_t\}$ and let $\mathbb{E}^y = \mathbb{E}^{(s,x)}$ denote the expectation w.r.t. R_y . In terms of Y_t the problem (39) can be written

$$g_0(x) = g^*(0, x) = \sup_{\tau} \mathbb{E}^{(0,x)}[g(Y_{\tau})] = \mathbb{E}^{(0,x)}[g(Y_{\tau^*})] \quad (42)$$

which is a special case of the problem

$$g^*(s, x) = \sup_{\tau} \mathbb{E}^{(s,x)}[g(Y_{\tau})] = \mathbb{E}^{(s,x)}[g(Y_{\tau^*})], \quad (43)$$

which is of the form (2)–(3) with X_t replaced by Y_t .

Note that the characteristic operator $\hat{\mathcal{A}}$ of Y_t is given by

$$\hat{\mathcal{A}}f(s, x) = \frac{\partial f}{\partial s}(s, x) + \mathcal{A}f(s, x); \quad f \in C^2(\mathbb{R} \times \mathbb{R}^n) \quad (44)$$

where $\mathcal{A}$ is the characteristic operator of X_t (working on the x -variables).

Example 2.1. Let $X_t = B_t$ be 1-dimensional Brownian motion and let the reward function be

$$g(t, \xi) = e^{-\alpha t + \beta \xi}; \quad \xi \in \mathbb{R}$$

where $\alpha, \beta \geq 0$ are constants. The characteristic operator $\hat{\mathcal{A}}$ of $Y_t^{(s,x)} = \begin{bmatrix} s+t \\ B_t^x \end{bmatrix}$ is given by

$$\hat{\mathcal{A}}f(s, x) = \frac{\partial f}{\partial s} + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}; \quad f \in C^2.$$

Thus

$$\hat{\mathcal{A}}g = \left(-\alpha + \frac{1}{2} \beta^2 \right) g,$$

so if $\beta^2 \leq 2\alpha$ then $g^* = g$ and the best policy is to stop immediately. If $\beta^2 > 2\alpha$ I have

$$U := \{(s, x); \hat{\mathcal{A}}g(s, x) > 0\} = \mathbb{R}^2$$

and therefore by (35) $D = \mathbb{R}^2$ and hence τ^* does not exist. If $\beta^2 > 2\alpha$ I can use Theorem 1.7 to prove that $g^* = \infty$:

$$\begin{aligned} \sup_{t \in S_n} \mathbb{E}^{(s,x)}[g(Y_t)] &= \sup_{t \in S_n} \mathbb{E}[e^{-\alpha(s+t)+\beta B_t^x}] \\ &= \sup_{t \in S_n} \left[e^{-\alpha(s+t)} \cdot e^{\beta x + \frac{1}{2} \beta^2 t} \right] \\ &= \sup_{t \in S_n} g(s, x) \cdot e^{(-\alpha + \frac{1}{2} \beta^2)t} = g(s, x) \cdot \exp \left(\left(-\alpha + \frac{1}{2} \beta^2 \right) 2^n \right), \end{aligned}$$

so $g_n(s, x) \rightarrow \infty$ as $n \rightarrow \infty$. Hence no optimal stopping exists in this case.

Example 2.2 (When is the right time to sell the stocks?). I now return to a specified optimal stopping problem: Suppose the price X_t at time t of a person's assets varies according to a stochastic differential equation of the form

$$dX_t = rX_t dt + \alpha X_t dB_t, \quad X_0 = x > 0,$$

where B_t is 1-dimensional Brownian motion and r, α are known constants. Suppose that connected to the sale of the assets there is a fixed fee/tax or transaction cost $a > 0$. Then if the person decides to sell at time t the discounted net of the sale is

$$e^{-\rho t}(X_t - a),$$

where $\rho > 0$ is a given discounting factor. The problem is to find a stopping time τ that maximizes

$$\mathbb{E}^{(s,x)}[e^{-\rho \tau}(X_\tau - a)] = \mathbb{E}^{(s,x)}[g(\tau, X_\tau)],$$

where

$$g(t, \xi) = e^{-\rho t}(\xi - a).$$

The characteristic operator $\hat{\mathcal{A}}$ of the process $Y_t = (s + t, X_t)$ is given by

$$\hat{\mathcal{A}}f(s, x) = \frac{\partial f}{\partial s} + rx \frac{\partial f}{\partial x} + \frac{1}{2} \alpha^2 x^2 \frac{\partial^2 f}{\partial x^2}; \quad f \in C^2(\mathbb{R}^2).$$

Hence $\hat{\mathcal{A}}g(s, x) = -\rho e^{-\rho s}(x - a) + rxe^{-\rho s} = e^{-\rho s}((r - \rho)x + \rho a)$. So

$$U := \{(s, x); \hat{\mathcal{A}}g(s, x) > 0\} = \begin{cases} \mathbb{R} \times \mathbb{R}^+ & \text{if } r \geq \rho \\ \{(s, x); x < \frac{a\rho}{\rho-r}\} & \text{if } r < \rho. \end{cases}$$

So if $r \geq \rho$ I have $U = D = \mathbb{R} \times \mathbb{R}^+$ so τ^* does not exist. If $r > \rho$ then $g^* = \infty$ while if $r = \rho$ then

$$g^*(s, x) = xe^{-\rho s}.$$

It remains to examine the case $r < \rho$. (If I regard ρ as the sum of interest rate, inflation and tax etc., this is not an unreasonable assumption in applications.) First I establish that the region D must be invariant w.r.t. t , in the sense that

$$D + (t_0, 0) = D \quad \text{for all } t_0. \quad (45)$$

To prove (45) consider

$$\begin{aligned} D + (t_0, 0) &= \{(t + t_0, x); (t, x) \in D\} = \{(s, x); (s - t_0, x) \in D\} \\ &= \{(s, x); g(s - t_0, x) < g^*(s - t_0, x)\} = \{(s, x); e^{\rho t_0} g(s, x) < e^{\rho t_0} g^*(s, x)\} \\ &= \{(s, x); g(s, x) < g^*(s, x)\} = D, \end{aligned}$$

where I have used that

$$\begin{aligned} g^*(s - t_0, x) &= \sup_{\tau} \mathbb{E}^{(s-t_0, x)}[e^{-\rho\tau}(X_{\tau} - a)] = \sup_{\tau} \mathbb{E}[e^{-\rho(\tau+s-t_0)}(X_{\tau}^x - a)] \\ &= e^{\rho t_0} \sup_{\tau} \mathbb{E}[e^{-\rho(\tau+s)}(X_{\tau}^x - a)] = e^{\rho t_0} g^*(s, x). \end{aligned}$$

Therefore the connected component of D that contains U must have the form

$$D(x_0) = \{(t, x); 0 < x < x_0\} \quad \text{for some } x_0 \geq \frac{a\rho}{\rho-r}.$$

Note that D cannot have any other components, for if V is a component of D disjoint from U then $\hat{\mathcal{A}}g < 0$ in V and so, if $y \in V$,

$$\mathbb{E}^y[g(Y_{\tau})] = g(y) + \mathbb{E}^y \left[\int_0^{\tau} \hat{\mathcal{A}}g(Y_t) dt \right] < g(y)$$

for all exit times τ bounded by the exit time from an x -bounded strip in V . From this I conclude by Theorem 1.9 c) that $g^*(y) = g(y)$, which implies $V = \emptyset$. Put $\tau(x_0) = \tau_{D(x_0)}$ and let me compute

$$\tilde{g}(s, x) = \tilde{g}_{x_0}(s, x) = \mathbb{E}^{(s, x)}[g(Y_{\tau(x_0)})]. \quad (46)$$

From the Dirichlet problem as worked out in Chapter 9 of Øksendal I know that $f = \tilde{g}$ is the (bounded) solution of the boundary value problem

$$\begin{cases} \frac{\partial f}{\partial s} + rx \frac{\partial f}{\partial x} + \frac{1}{2} \alpha^2 x^2 \frac{\partial^2 f}{\partial x^2} = 0 & \text{for } 0 < x < x_0 \\ f(s, x_0) = e^{-\rho s}(x_0 - a). \end{cases} \quad (47)$$

(Note that $\mathbb{R} \times \{0\}$ does not contain any regular boundary points of D w.r.t. $Y_t = (s+t, X_t)$.) If I try a solution of (47) of the form

$$f(s, x) = e^{-\rho s} \phi(x)$$

I get the following 1-dimensional problem

$$\begin{cases} -\rho\phi + rx\phi'(x) + \frac{1}{2}\alpha^2 x^2 \phi''(x) = 0 & \text{for } 0 < x < x_0 \\ \phi(x_0) = x_0 - a. \end{cases} \quad (48)$$

The general solution ϕ of (48) is

$$\phi(x) = C_1 x^{\gamma_1} + C_2 x^{\gamma_2},$$

where C_1, C_2 are arbitrary constants and

$$\gamma_i = \alpha^{-2} \left[\frac{1}{2}\alpha^2 - r \pm \sqrt{\left(r - \frac{1}{2}\alpha^2\right)^2 + 2\rho\alpha^2} \right] \quad (i = 1, 2), \quad \gamma_2 < 0 < \gamma_1.$$

Since $\phi(x)$ is bounded as $x \rightarrow 0$ I must have $C_2 = 0$ and the boundary requirement $\phi(x_0) = x_0 - a$ gives $C_1 = x_0^{-\gamma_1}(x_0 - a)$. I conclude that the bounded solution f of (47) is

$$\tilde{g}_{x_0}(s, x) = f(s, x) = e^{-\rho s} (x_0 - a) \left(\frac{x}{x_0} \right)^{\gamma_1}. \quad (49)$$

If I fix (s, x) then the value of x_0 which maximizes $\tilde{g}_{x_0}(s, x)$ is easily seen to be given by

$$x_0 = x_{\max} = \frac{a\gamma_1}{\gamma_1 - 1} \quad (50)$$

(note that $\gamma_1 > 1$ if and only if $r < \rho$). Thus I have arrived at the candidate $\tilde{g}_{x_{\max}}(s, x)$ for $g^*(s, x) = \sup_{\tau} \mathbb{E}^{(s, x)}[e^{-\rho\tau}(X_{\tau} - a)]$. To verify that I indeed have $\tilde{g}_{x_{\max}} = g^*$ it would suffice to prove that $\tilde{g}_{x_{\max}}$ is a supermeanvalued majorant of g (see Corollary 1.10). This can be done, but I do not give the details here, since this problem can be solved more easily by Theorem 4.1.

The conclusion is therefore that one should sell the assets the first time the price of them reaches the value $x_{\max} = \frac{a\gamma_1}{\gamma_1 - 1}$. The expected discounted profit obtained from this strategy is

$$g^*(s, x) = \tilde{g}_{x_{\max}}(s, x) = e^{-\rho s} \left(\frac{\gamma_1 - 1}{a} \right)^{\gamma_1 - 1} \left(\frac{x}{\gamma_1} \right)^{\gamma_1}.$$

Remark. The reader is invited to check that the value $x_0 = x_{\max}$ is the only value of x_0 which makes the function

$$x \mapsto \tilde{g}_{x_0}(s, x) \quad (\text{given by (46)})$$

continuously differentiable at x_0 . This is not a coincidence. In fact, it illustrates a general phenomenon which is known as the high contact (or smooth fit) principle.

3 Optimal Stopping Problems Involving an Integral

Let

$$dY_t = b(Y_t)dt + \sigma(Y_t)dB_t, \quad Y_0 = y \quad (51)$$

be an Itô diffusion in $\mathbb{R}^k$. Let $g : \mathbb{R}^k \rightarrow \mathbb{R}^+$ be continuous and let $f : \mathbb{R}^k \rightarrow \mathbb{R}^+$ be Lipschitz continuous with at most linear growth. (These conditions can be relaxed. See (37) and Theorem 4.1.) Consider the optimal stopping problem: Find $G^*(y)$ and τ^* such that

$$G^*(y) = \sup_{\tau} \mathbb{E}^y \left[\int_0^{\tau} f(Y_t)dt + g(Y_{\tau}) \right] = \mathbb{E}^y \left[\int_0^{\tau^*} f(Y_t)dt + g(Y_{\tau^*}) \right]. \quad (52)$$

This problem can be reduced to my original problem (2)–(3) by proceeding as follows: Define the Itô diffusion Z_t in $\mathbb{R}^k \times \mathbb{R} = \mathbb{R}^{k+1}$ by

$$dZ_t = \begin{bmatrix} dY_t \\ dW_t \end{bmatrix} := \begin{bmatrix} b(Y_t) \\ f(Y_t) \end{bmatrix} dt + \begin{bmatrix} \sigma(Y_t) \\ 0 \end{bmatrix} dB_t; \quad Z_0 = z = (y, w). \quad (53)$$

Then I see that

$$G^*(y) = \sup_{\tau} \mathbb{E}^{(y,0)} [W_{\tau} + g(Y_{\tau})] = \sup_{\tau} \mathbb{E}^{(y,0)} [G(Z_{\tau})] \quad (54)$$

with

$$G(z) := G(y, w) := g(y) + w; \quad z = (y, w) \in \mathbb{R}^k \times \mathbb{R}. \quad (55)$$

This is again a problem of the type (2)–(3) with X_t replaced by Z_t and g replaced by G . Note that the connection between the characteristic operators $\mathcal{A}_Y$ of Y_t and $\mathcal{A}_Z$ of Z_t is given by

$$\mathcal{A}_Z \phi(z) = \mathcal{A}_Z \phi(y, w) = \mathcal{A}_Y \phi(y, w) + f(y) \frac{\partial \phi}{\partial w}, \quad \phi \in C^2(\mathbb{R}^{k+1}). \quad (56)$$

In particular, if $G(y, w) = g(y) + w \in C^2(\mathbb{R}^{k+1})$ then

$$\mathcal{A}_Z G(y, w) = \mathcal{A}_Y g(y) + f(y). \quad (57)$$

Example 3.1. Consider the optimal stopping problem

$$\gamma(x) = \sup_{\tau} \mathbb{E}^x \left[\int_0^{\tau} \theta e^{-\rho t} X_t dt + e^{-\rho \tau} X_{\tau} \right],$$

where

$$dX_t = \alpha X_t dt + \beta X_t dB_t; \quad X_0 = x > 0$$

is geometric Brownian motion (α, β, θ constants, $\theta > 0$). I put

$$dY_t = \begin{bmatrix} dt \\ dX_t \end{bmatrix} = \begin{bmatrix} 1 \\ \alpha X_t \end{bmatrix} dt + \begin{bmatrix} 0 \\ \beta X_t \end{bmatrix} dB_t; \quad Y_0 = (s, x)$$

and

$$dZ_t = \begin{bmatrix} dY_t \\ dW_t \end{bmatrix} = \begin{bmatrix} 1 \\ \alpha X_t \\ \theta e^{-\rho t} X_t \end{bmatrix} dt + \begin{bmatrix} 0 \\ \beta X_t \\ 0 \end{bmatrix} dB_t; \quad Z_0 = (s, x, w).$$

Then with

$$f(y) = f(s, x) = \theta e^{-\rho s} x, \quad g(y) = e^{-\rho s} x$$

and

$$G(s, x, w) = g(s, x) + w = e^{-\rho s} x + w$$

I have

$$\mathcal{A}_Z G = \frac{\partial G}{\partial s} + \alpha x \frac{\partial G}{\partial x} + \frac{1}{2} \beta^2 x^2 \frac{\partial^2 G}{\partial x^2} + \theta e^{-\rho s} x \frac{\partial G}{\partial w} = (-\rho + \alpha + \theta) e^{-\rho s} x.$$

Hence

$$U = \{(s, x, w); \mathcal{A}_Z G(s, x, w) > 0\} = \begin{cases} \mathbb{R}^3 & \text{if } \rho < \alpha + \theta \\ \emptyset & \text{if } \rho \geq \alpha + \theta. \end{cases}$$

From this I conclude:

If $\rho \geq \alpha + \theta$ then $\tau^* = 0$

$$\text{and } G^*(s, x, w) = G(s, x, w) = e^{-\rho s} x + w. \quad (58)$$

If $\alpha < \rho < \alpha + \theta$ then τ^* does not exist

$$\text{and } G^*(s, x, w) = \frac{\theta x}{\rho - \alpha} e^{-\rho s} + w. \quad (59)$$

$$\text{If } \rho \leq \alpha \text{ then } \tau^* \text{ does not exist and } G^* = \infty. \quad (60)$$

4 Connection with Variational Inequalities

The ‘high contact principle’ says, roughly, that – under certain conditions – the solution g^* of (2)–(3) is a C^1 function on $\mathbb{R}^n$ if $g \in C^2(\mathbb{R}^n)$. This is a useful information which can help me to determine g^* . Indeed, this principle is so useful that it is frequently applied in the literature also in cases where its validity has not been rigorously proved.

Fortunately it turns out to be easy to prove a sufficiency condition of high contact type, i.e. a kind of verification theorem for optimal stopping, which makes it easy to verify that a given candidate for g^* (that I may have found by guessing or intuition) is actually equal to g^* . The result below is a simplified variant of a result in Brekke and Øksendal (1991):

In the following I fix a domain V in $\mathbb{R}^k$ and I let

$$dY_t = b(Y_t)dt + \sigma(Y_t)dB_t; \quad Y_0 = y \quad (61)$$

be an Itô diffusion in $\mathbb{R}^k$. Define

$$T = T(y, \omega) = \inf\{t > 0; Y_t(\omega) \notin V\}. \quad (62)$$

Let $f : \mathbb{R}^k \rightarrow \mathbb{R}$, $g : \mathbb{R}^k \rightarrow \mathbb{R}$ be continuous functions satisfying

$$(a) \quad \mathbb{E}^y \left[\int_0^T |f(Y_t)| dt \right] < \infty \quad \text{for all } y \in \mathbb{R}^k \quad (63)$$

and

$$(b) \quad \text{the family } \{g^-(Y_\tau); \tau \text{ stopping time}, \tau \leq T\} \text{ is uniformly integrable w.r.t. } R_y, \quad (64)$$

for all $y \in \mathbb{R}^k$.

Consider the following problem: Find $\Phi(y)$ and $\tau^* \leq T$ such that

$$\Phi(y) = \sup_{\tau \leq T} J^\tau(y) = J^{\tau^*}(y), \quad (65)$$

where

$$J^\tau(y) = \mathbb{E}^y \left[\int_0^\tau f(Y_t) dt + g(Y_\tau) \right] \quad \text{for } \tau \leq T.$$

Note that since $J^0(y) = g(y)$ I have

$$\Phi(y) \geq g(y) \quad \text{for all } y \in V. \quad (66)$$

I can now formulate the variational inequalities. As usual I let

$$L = L_Y = \sum_{i=1}^k b_i(y) \frac{\partial}{\partial y_i} + \frac{1}{2} \sum_{i,j=1}^k (\sigma \sigma^T)_{ij}(y) \frac{\partial^2}{\partial y_i \partial y_j}$$

be the partial differential operator which coincides with the generator $\mathcal{A}_Y$ of Y_t on $C_0^2(\mathbb{R}^k)$.

Theorem 4.1 (Variational inequalities for optimal stopping). *Suppose I can find a function $\phi : V \rightarrow \mathbb{R}$ such that*

- (i) $\phi \in C^1(V) \cap C(V)$
- (ii) $\phi \geq g$ on V and $\phi = g$ on ∂V .

Define

$$D = \{x \in V; \phi(x) > g(x)\}.$$

Suppose Y_t spends 0 time on ∂D a.s., i.e.

$$(iii) \quad \mathbb{E}^y \left[\int_0^T \chi_{\partial D}(Y_t) dt \right] = 0 \quad \text{for all } y \in V$$

and suppose that

- (iv) ∂D is a Lipschitz surface, i.e. ∂D is locally the graph of a function $h : \mathbb{R}^{k-1} \rightarrow \mathbb{R}$ such that there exists $K < \infty$ with

$$|h(x) - h(y)| \leq K|x - y| \quad \text{for all } x, y.$$

Moreover, suppose the following:

(v) $\phi \in C^2(V \setminus \partial D)$ and the second order derivatives of ϕ are locally bounded near ∂D

(vi) $L\phi + f \leq 0$ on $V \setminus D$

(vii) $L\phi + f = 0$ on D

(viii) $\tau_D := \inf\{t > 0; Y_t \notin D\} < \infty$ a.s. R_y for all $y \in V$

and

(ix) the family $\{\phi(Y_\tau); \tau \leq \tau_D\}$ is uniformly integrable w.r.t. R_y , for all $y \in V$.

Then

$$\phi(y) = \Phi(y) = \sup_{\tau \leq T} \mathbb{E}^y \left[\int_0^\tau f(Y_t) dt + g(Y_\tau) \right]; \quad y \in V \quad (67)$$

and

$$\tau^* = \tau_D \quad (68)$$

is an optimal stopping time for this problem.

Proof. By (i), (iv) and (v) I can find a sequence of functions $\phi_j \in C^2(V) \cap C(V)$, $j = 1, 2, \dots$, such that

(a) $\phi_j \rightarrow \phi$ uniformly on compact subsets of V , as $j \rightarrow \infty$

(b) $L\phi_j \rightarrow L\phi$ uniformly on compact subsets of $V \setminus \partial D$, as $j \rightarrow \infty$

(c) $\{L\phi_j\}_{j=1}^\infty$ is locally bounded on V .

For $R > 0$ put $T_R = \min(R, \inf\{t > 0; |Y_t| \geq R\})$ and let $\tau \leq T$ be a stopping time. Let $y \in V$. Then by Dynkin's formula

$$\mathbb{E}^y[\phi_j(Y_{\tau \wedge T_R})] = \phi_j(y) + \mathbb{E}^y \left[\int_0^{\tau \wedge T_R} L\phi_j(Y_t) dt \right]. \quad (69)$$

Hence by (a), (b), (c) and (iii) and the Fatou lemma

$$\begin{aligned} \phi(y) &= \lim_{j \rightarrow \infty} \mathbb{E}^y \left[\int_0^{\tau \wedge T_R} -L\phi_j(Y_t) dt + \phi_j(Y_{\tau \wedge T_R}) \right] \\ &\geq \mathbb{E}^y \left[\int_0^{\tau \wedge T_R} -L\phi(Y_t) dt + \phi(Y_{\tau \wedge T_R}) \right]. \end{aligned} \quad (70)$$

Therefore, by (ii), (iii), (vi) and (vii),

$$\phi(y) \geq \mathbb{E}^y \left[\int_0^{\tau \wedge T_R} f(Y_t) dt + g(Y_{\tau \wedge T_R}) \right].$$

Hence by the Fatou lemma and (63), (64)

$$\begin{aligned}\phi(y) &\geq \lim_{R \rightarrow \infty} \mathbb{E}^y \left[\int_0^{\tau \wedge T_R} f(Y_t) dt + g(Y_{\tau \wedge T_R}) \right] \\ &\geq \mathbb{E}^y \left[\int_0^{\tau} f(Y_t) dt + g(Y_{\tau}) \right].\end{aligned}$$

Since $\tau \leq T$ was arbitrary, I conclude that

$$\phi(y) \geq \Phi(y) \quad \text{for all } y \in V. \quad (71)$$

If $y \notin D$ then $\phi(y) = g(y) \leq \Phi(y)$ so by (71) I have

$$\phi(y) = \Phi(y) \text{ and } \hat{\tau} = \hat{\tau}(y, \omega) := 0 \text{ is optimal for } y \notin D. \quad (72)$$

Next, suppose $y \in D$. Let $\{D_k\}_{k=1}^{\infty}$ be an increasing sequence of open sets D_k such that $D_k \subset D$, D_k is compact and $D = \bigcup_{k=1}^{\infty} D_k$. Put $\tau_k = \inf\{t > 0; Y_t \notin D_k\}$, $k = 1, 2, \dots$. By Dynkin's formula I have for $y \in D_k$,

$$\begin{aligned}\phi(y) &= \lim_{j \rightarrow \infty} \phi_j(y) = \lim_{j \rightarrow \infty} \mathbb{E}^y \left[\int_0^{\tau_k \wedge T_R} -L\phi_j(Y_t) dt + \phi_j(Y_{\tau_k \wedge T_R}) \right] \\ &= \mathbb{E}^y \left[\int_0^{\tau_k \wedge T_R} -L\phi(Y_t) dt + \phi(Y_{\tau_k \wedge T_R}) \right] \\ &= \mathbb{E}^y \left[\int_0^{\tau_k \wedge T_R} f(Y_t) dt + \phi(Y_{\tau_k \wedge T_R}) \right].\end{aligned}$$

So by uniform integrability and (ii), (vii), (viii) I get

$$\begin{aligned}\phi(y) &= \lim_{R, k \rightarrow \infty} \mathbb{E}^y \left[\int_0^{\tau_k \wedge T_R} f(Y_t) dt + \phi(Y_{\tau_k \wedge T_R}) \right] \\ &= \mathbb{E}^y \left[\int_0^{\tau_D} f(Y_t) dt + g(Y_{\tau_D}) \right] \\ &= J^{\tau_D}(y) \leq \Phi(y).\end{aligned} \quad (73)$$

Combining (71) and (73) I get

$$\phi(y) \geq \Phi(y) \geq J^{\tau_D}(y) = \phi(y)$$

so

$$\phi(y) = \Phi(y) \text{ and } \hat{\tau}(y, \omega) := \tau_D \text{ is optimal when } y \in D. \quad (74)$$

From (72) and (74) I conclude that

$$\phi(y) = \Phi(y) \quad \text{for all } y \in V.$$

Moreover, the stopping time $\hat{\tau}$ defined by

$$\hat{\tau}(y, \omega) = \begin{cases} 0 & \text{for } y \notin D \\ \tau_D & \text{for } y \in D \end{cases}$$

is optimal. By Theorem 1.12 I conclude that τ_D is optimal also. $\square$

Example 4.2. *To illustrate Theorem 4.1 let me apply it to reconsider Example 2.2: Rather than proving (45) and the following properties of D , I now simply guess/assume that D has the form*

$$D = \{(s, x); 0 < x < x_0\}$$

for some $x_0 > 0$, which is intuitively reasonable. Then I solve (48) for arbitrary x_0 and I arrive at the following candidate ϕ for g^ :*

$$\phi(s, x) = \begin{cases} e^{-\rho s}(x_0 - a) \left(\frac{x}{x_0}\right)^{\gamma_1} & \text{for } 0 < x < x_0 \\ e^{-\rho s}(x - a) & \text{for } x \geq x_0. \end{cases}$$

The requirement that $\phi \in C^1$ (Theorem 4.1 (i)) gives the value (50) for x_0 . It is clear that $\phi \in C^2$ outside ∂D and by construction $L\phi = 0$ on D . Moreover, conditions (iii), (iv), (viii) and (ix) clearly hold. It remains to verify that (ii) $\phi(s, x) > g(s, x)$ for $0 < x < x_0$ and (vi) $L\phi + f \leq 0$ on $V \setminus D$, i.e. $L(e^{-\rho s}(x - a)) \leq 0$ for $x > x_0$, i.e., $Lg(s, x) \leq 0$ for $x > x_0$. This is easily done by direct calculation (assuming $r < \rho$).

I conclude that $\phi = g^$ and $\tau^* = \tau_D$ is optimal (with the value (50) for x_0).*

5 Selected Exercises

Exercise 9

Problem Statement: Solve the optimal stopping problem

$$\gamma(x) = \sup_{\tau} \mathbb{E}^x \left[\int_0^{\tau} e^{-\rho t} B_t^2 dt + e^{-\rho \tau} B_{\tau}^2 \right]$$

where B_t is a standard 1-dimensional Brownian motion, and $\rho > 0$ is a constant discount factor.

Solution:

I am dealing with an optimal stopping problem involving a running integral cost. To reduce this to the standard form without integrals, I introduce a dummy variable W_t to track the accumulated running reward:

$$W_t = \int_0^t e^{-\rho s} B_s^2 ds \implies dW_t = e^{-\rho t} B_t^2 dt$$

I augment my state space to a 3-dimensional Markov process $\mathbf{Z}_t = (t, B_t, W_t)$ starting at $(0, x, 0)$. My total reward function evaluated at the stopping time τ is:

$$G(t, x, w) = w + e^{-\rho t} x^2$$

The problem is now to find $\sup_{\tau} \mathbb{E}^{(0, x, 0)}[G(\mathbf{Z}_{\tau})]$. By the formulation, the characteristic operator $\mathcal{A}_Z$ applied to my reward function G is:

$$\mathcal{A}_Z G(t, x, w) = \frac{\partial G}{\partial t} + \frac{1}{2} \frac{\partial^2 G}{\partial x^2} + (e^{-\rho t} x^2) \frac{\partial G}{\partial w}$$

Substituting $G(t, x, w)$:

$$\begin{aligned} \frac{\partial G}{\partial t} &= -\rho e^{-\rho t} x^2 \\ \frac{\partial^2 G}{\partial x^2} &= 2e^{-\rho t} \\ \frac{\partial G}{\partial w} &= 1 \end{aligned}$$

$$\mathcal{A}_Z G(t, x, w) = -\rho e^{-\rho t} x^2 + e^{-\rho t} + e^{-\rho t} x^2 = e^{-\rho t} [1 + (1 - \rho)x^2]$$

The sign of the immediate drift is entirely determined by the term $[1 + (1 - \rho)x^2]$. I define U as the region where the expected reward is strictly increasing ($\mathcal{A}_Z G > 0$). This leads to two distinct cases depending on the discount factor ρ .

Case 1: $\rho \leq 1$ (Weak Discounting)

If $\rho \leq 1$, then $(1 - \rho) \geq 0$. Because $x^2 \geq 0$, I have:

$$1 + (1 - \rho)x^2 \geq 1 > 0 \quad \text{for all } x \in \mathbb{R}$$

Because the characteristic operator is strictly positive everywhere, the expected reward increases indefinitely the longer I wait. The region U is the entire state space $\mathbb{R}$. Therefore, **an optimal stopping time τ^* does not exist**, and I should never stop. I evaluate the supremum by taking the limit as $\tau \rightarrow \infty$:

$$\gamma(x) = \mathbb{E}^x \left[\int_0^\infty e^{-\rho t} B_t^2 dt + \lim_{t \rightarrow \infty} e^{-\rho t} B_t^2 \right]$$

Since $\mathbb{E}^x[B_t^2] = x^2 + t$, the boundary term $\lim_{t \rightarrow \infty} e^{-\rho t}(x^2 + t)$ vanishes. The expected reward is purely the integral:

$$\gamma(x) = \int_0^\infty e^{-\rho t}(x^2 + t)dt = x^2 \int_0^\infty e^{-\rho t} dt + \int_0^\infty t e^{-\rho t} dt$$

$$\gamma(x) = \frac{x^2}{\rho} + \frac{1}{\rho^2}$$

Case 2: $\rho > 1$ (Strong Discounting)

If $\rho > 1$, then $(1 - \rho) < 0$. The region U where the drift is positive is given by:

$$1 - (\rho - 1)x^2 > 0 \quad \implies \quad |x| < \frac{1}{\sqrt{\rho - 1}}$$

Since the process is symmetric and the boundary of U is independent of time, I guess that the continuation region D takes the form $D = (-x_0, x_0)$ for some unknown boundary $x_0 \geq \frac{1}{\sqrt{\rho - 1}}$.

Inside the continuation region D , the value function $V(x)$ must satisfy the martingale property. Removing the explicit time-dependence $e^{-\rho t}$, this yields the standard free-boundary ODE:

$$\frac{1}{2}V''(x) - \rho V(x) + x^2 = 0$$

The general solution to this linear ODE is the sum of the homogeneous solution and a particular solution.

1. **Homogeneous:** $\frac{1}{2}V_h''(x) - \rho V_h(x) = 0$. The roots are $\pm\sqrt{2\rho}$. Due to the symmetry of Brownian motion, $V(x) = V(-x)$, meaning the coefficients must be equal. Thus, $V_h(x) = C \cosh(x\sqrt{2\rho})$.
2. **Particular:** Guessing a quadratic form $V_p(x) = Ax^2 + B$ and substituting it into the ODE gives $A = \frac{1}{\rho}$ and $B = \frac{1}{\rho^2}$.

Therefore, my candidate value function inside D is:

$$V(x) = C \cosh(x\sqrt{2\rho}) + \frac{x^2}{\rho} + \frac{1}{\rho^2}$$

Applying Boundary Conditions (Smooth Pasting): At the optimal boundary x_0 , the value function must match the stopping payout $g(x) = x^2$ (Value Matching) and their first derivatives must be equal (Smooth Pasting).

1. **Value Matching:** $V(x_0) = x_0^2$

$$C \cosh(x_0\sqrt{2\rho}) + \frac{x_0^2}{\rho} + \frac{1}{\rho^2} = x_0^2 \quad \implies \quad C \cosh(x_0\sqrt{2\rho}) = x_0^2 \left(\frac{\rho - 1}{\rho} \right) - \frac{1}{\rho^2}$$

2. **Smooth Pasting:** $V'(x_0) = 2x_0$

$$C\sqrt{2\rho}\sinh(x_0\sqrt{2\rho}) + \frac{2x_0}{\rho} = 2x_0 \implies C\sqrt{2\rho}\sinh(x_0\sqrt{2\rho}) = 2x_0\left(\frac{\rho-1}{\rho}\right)$$

To isolate x_0 , I divide the smooth pasting equation by the value matching equation. This eliminates the constant C :

$$\frac{C\sqrt{2\rho}\sinh(x_0\sqrt{2\rho})}{C\cosh(x_0\sqrt{2\rho})} = \frac{2x_0\left(\frac{\rho-1}{\rho}\right)}{x_0^2\left(\frac{\rho-1}{\rho}\right) - \frac{1}{\rho^2}}$$

Multiplying the numerator and denominator of the right side by ρ^2 yields the final transcendental equation for the free boundary x_0 :

$$\sqrt{2\rho}\tanh(x_0\sqrt{2\rho}) = \frac{2x_0\rho(\rho-1)}{x_0^2\rho(\rho-1) - 1}$$

Because $\rho > 1$, this equation has a unique positive root x_0 . Once x_0 is computed, the constant C is uniquely determined by plugging x_0 back into the Value Matching equation.

Final Conclusion for $\rho > 1$:

The optimal stopping time is $\tau^* = \inf\{t > 0 : |B_t| \geq x_0\}$, and the value function is:

$$\gamma(x) = \begin{cases} C\cosh(x\sqrt{2\rho}) + \frac{x^2}{\rho} + \frac{1}{\rho^2} & \text{if } |x| < x_0 \\ x^2 & \text{if } |x| \geq x_0 \end{cases}$$

Exercise 10

Problem Statement: Prove the following simple, but useful, observation: Let $W = \{(s, x) : \exists \tau \text{ with } g(s, x) < \mathbb{E}^{(s, x)}[g(s + \tau, X_\tau)]\}$. Prove that $W \subset D$.

Solution:

This observation relies directly on the definition of the optimal value function g^* as the supremum over all possible stopping times.

Recall the definition of the continuation region D :

$$D = \{(s, x) : g(s, x) < g^*(s, x)\}$$

where $g^*(s, x) = \sup_{\tau} \mathbb{E}^{(s, x)}[g(s + \tau, X_\tau)]$.

Let (s_0, x_0) be an arbitrary point in W . By the definition of the set W , there exists *at least one* specific stopping time, let me call it τ_1 , such that:

$$g(s_0, x_0) < \mathbb{E}^{(s_0, x_0)}[g(s_0 + \tau_1, X_{\tau_1})]$$

Because the optimal value $g^*(s_0, x_0)$ is the supremum over the expected rewards of *all* possible stopping times, it must be greater than or equal to the expected reward of my specific strategy τ_1 :

$$\mathbb{E}^{(s_0, x_0)}[g(s_0 + \tau_1, X_{\tau_1})] \leq \sup_{\tau} \mathbb{E}^{(s_0, x_0)}[g(s_0 + \tau, X_\tau)] = g^*(s_0, x_0)$$

Chaining these two inequalities together, I get:

$$g(s_0, x_0) < g^*(s_0, x_0)$$

By definition, this strict inequality means that $(s_0, x_0) \in D$. Since this holds true for any arbitrary point in W , I conclude that $W \subset D$.

Exercise 11

Problem Statement: Consider the optimal stopping problem

$$g^*(s, x) = \sup_{\tau} \mathbb{E}^{(s, x)} [e^{-\rho(s+\tau)} B_{\tau}^+]$$

where $B_t \in \mathbf{R}$ and $x^+ = \max\{x, 0\}$. $\rho > 0$ is a constant discount factor. a) Prove that the continuation region D has the form $D = \{(s, x); x < x_0\}$ for some $x_0 > 0$. b) Determine x_0 and find g^* . c) Verify the high contact principle.

Solution:

a) Form of the Continuation Region D

I define the time-space process $\mathbf{Y}_t = (s + t, B_t)$ starting at (s, x) . My reward function is $g(s, x) = e^{-\rho s} x^+$.

First, I establish that the continuation region $D = \{(s, x) : g(s, x) < g^*(s, x)\}$ is time-invariant. Shifting the starting time by an arbitrary constant t_0 :

$$g^*(s - t_0, x) = \sup_{\tau} \mathbb{E}^{(s-t_0, x)} [e^{-\rho(s-t_0+\tau)} B_{\tau}^+]$$

Because the transition probabilities of Brownian motion are independent of the starting time, I can factor out the constant exponential shift:

$$g^*(s - t_0, x) = e^{\rho t_0} \sup_{\tau} \mathbb{E}^{(s, x)} [e^{-\rho(s+\tau)} B_{\tau}^+] = e^{\rho t_0} g^*(s, x)$$

Similarly, the immediate payoff satisfies $g(s - t_0, x) = e^{\rho t_0} g(s, x)$. Therefore, the inequality $g(s, x) < g^*(s, x)$ holds if and only if $g(s - t_0, x) < g^*(s - t_0, x)$. This proves the boundary of D depends only on the spatial coordinate x .

Next, I evaluate the optimal strategy for $x \leq 0$. If $x \leq 0$, the immediate payoff is $g(s, x) = 0$. However, because Brownian motion has infinite variation, if I wait for any strictly positive time $\tau > 0$, there is a strictly positive probability that $B_{\tau} > 0$, yielding a positive discounted reward. Thus, for all $x \leq 0$:

$$g^*(s, x) > 0 = g(s, x)$$

This implies that the entire left half-plane $(-\infty, 0]$ is strictly inside the continuation region D . Since D is time-invariant and must be a connected component anchored at $-\infty$, D must take the form:

$$D = \{(s, x); x < x_0\}$$

for some strictly positive boundary $x_0 > 0$.

b) Determining x_0 and the Value Function g^*

By the Dirichlet problem as worked out in Chapter 9 of Øksendal, if I restrict myself to the continuation region D , the optimal expected reward $f(s, x) = g^*(s, x)$ must be the bounded solution to the Dirichlet boundary value problem. Specifically, f must be harmonic with respect to the characteristic operator $\hat{\mathcal{A}}$ of $\mathbf{Y}_t$:

$$\begin{aligned}\hat{\mathcal{A}}f &= 0 \quad \text{for } x < x_0 \\ f(s, x_0) &= g(s, x_0) = e^{-\rho s} x_0\end{aligned}$$

The characteristic operator for $\mathbf{Y}_t = (s + t, B_t)$ is:

$$\hat{\mathcal{A}} = \frac{\partial}{\partial s} + \frac{1}{2} \frac{\partial^2}{\partial x^2}$$

Setting $\hat{\mathcal{A}}f = 0$, I get my PDE:

$$\frac{\partial f}{\partial s} + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} = 0$$

Because the boundary condition has the separable form $e^{-\rho s} x_0$, I guess a solution of the form $f(s, x) = e^{-\rho s} \phi(x)$. Substituting this into the PDE yields the 1-dimensional ODE:

$$-\rho e^{-\rho s} \phi(x) + \frac{1}{2} e^{-\rho s} \phi''(x) = 0 \quad \implies \quad \phi''(x) - 2\rho \phi(x) = 0$$

The general solution to this linear ODE is:

$$\phi(x) = C_1 e^{x\sqrt{2\rho}} + C_2 e^{-x\sqrt{2\rho}}$$

I now apply my spatial boundary conditions to find the constants: 1. **As $x \rightarrow -\infty$:** The expected discounted payoff of a Brownian motion starting infinitely far away from the positive region must be 0. To prevent the function from blowing up to infinity as x becomes large and negative, I must enforce $C_2 = 0$. This leaves $\phi(x) = C_1 e^{x\sqrt{2\rho}}$. 2. **At $x = x_0$:** The function must match the immediate payoff $\phi(x_0) = x_0$.

$$C_1 e^{x_0\sqrt{2\rho}} = x_0 \quad \implies \quad C_1 = x_0 e^{-x_0\sqrt{2\rho}}$$

This gives me a family of candidate value functions, parameterized by the choice of boundary x_0 :

$$\tilde{g}^{x_0}(s, x) = e^{-\rho s} x_0 e^{(x-x_0)\sqrt{2\rho}}$$

To find the globally optimal value function g^* , I must choose the boundary x_0 that maximizes this expected reward. I maximize the spatial component $H(x_0) = x_0 e^{-x_0\sqrt{2\rho}}$ by setting its derivative to 0:

$$H'(x_0) = e^{-x_0\sqrt{2\rho}} - x_0 \sqrt{2\rho} e^{-x_0\sqrt{2\rho}} = 0$$

Factoring out the exponential, I get:

$$1 - x_0 \sqrt{2\rho} = 0 \implies x_0 = \frac{1}{\sqrt{2\rho}}$$

Substituting this optimal boundary back into my candidate function, I obtain the final value function:

$$g^*(s, x) = \begin{cases} e^{-\rho s} \frac{1}{\sqrt{2\rho}} e^{(x\sqrt{2\rho}-1)} & \text{for } x < \frac{1}{\sqrt{2\rho}} \\ e^{-\rho s} x & \text{for } x \geq \frac{1}{\sqrt{2\rho}} \end{cases}$$

c) Verifying the High Contact Principle

The high contact (smooth pasting) principle dictates that at the optimal free boundary, the first spatial derivative of the value function must perfectly match the first spatial derivative of the immediate payoff function.

I evaluate both derivatives precisely at the boundary (s, x_0) .

1. **Derivative of the immediate payoff g :** For $x > 0$, $g(s, x) = e^{-\rho s} x$.

$$\frac{\partial g}{\partial x} = e^{-\rho s}$$

2. **Derivative of the optimal value function g^* :** Approaching from inside the continuation region D :

$$g^*(s, x) = e^{-\rho s} x_0 e^{(x-x_0)\sqrt{2\rho}}$$

$$\frac{\partial g^*}{\partial x} = e^{-\rho s} x_0 \sqrt{2\rho} e^{(x-x_0)\sqrt{2\rho}}$$

Evaluating this derivative exactly at $x = x_0$:

$$\left. \frac{\partial g^*}{\partial x} \right|_{x=x_0} = e^{-\rho s} x_0 \sqrt{2\rho} e^0 = e^{-\rho s} x_0 \sqrt{2\rho}$$

Finally, I substitute my calculated optimal boundary $x_0 = \frac{1}{\sqrt{2\rho}}$ into this expression:

$$\left. \frac{\partial g^*}{\partial x} \right|_{x=x_0} = e^{-\rho s} \left(\frac{1}{\sqrt{2\rho}} \right) \sqrt{2\rho} = e^{-\rho s}$$

Thus, $\frac{\partial g^*}{\partial x} = \frac{\partial g}{\partial x}$ at (s, x_0) . The high contact principle holds perfectly.

Exercise 12

Problem Statement: The problem is to find the optimal time τ^* to sell an asset to maximize the expected discounted reward. The reward function is given by:

$$g(t, \xi) = e^{-\rho t} (\xi - 1)^+$$

The asset price X_t follows a geometric Brownian motion:

$$dX_t = rX_t dt + \alpha X_t dB_t, \quad X_0 = x > 0$$

where I assume $r < \rho$. Find the continuation region, boundary value problem, and determine x_0 using the high contact principle.

Solution:

a) Form of the Continuation Region D

I define the time-space process $\mathbf{Y}_t = (s + t, X_t)$ starting at (s, x) . My optimal expected reward is $g^*(s, x)$.

First, I establish that the continuation region $D = \{(s, x) : g(s, x) < g^*(s, x)\}$ is invariant with respect to time. Shifting the starting time by a constant t_0 :

$$g^*(s - t_0, x) = \sup_{\tau} \mathbb{E}^{(s-t_0, x)} \left[e^{-\rho(s-t_0+\tau)} (X_{\tau} - 1)^+ \right]$$

Because the dynamics of geometric Brownian motion are independent of the calendar time, I can factor out the constant exponential term:

$$g^*(s - t_0, x) = e^{\rho t_0} \sup_{\tau} \mathbb{E}^{(s, x)} \left[e^{-\rho(s+\tau)} (X_{\tau} - 1)^+ \right] = e^{\rho t_0} g^*(s, x)$$

Since the immediate payoff also shifts as $g(s - t_0, x) = e^{\rho t_0} g(s, x)$, the strict inequality $g < g^*$ is perfectly preserved across time. Thus, the boundary of D depends solely on the price x .

Next, I identify the region U where the characteristic operator of the payoff is strictly positive ($\hat{\mathcal{A}}g > 0$). For $x > 1$, the payoff is $g(s, x) = e^{-\rho s}(x - 1)$. Applying the characteristic operator $\hat{\mathcal{A}} = \frac{\partial}{\partial s} + rx \frac{\partial}{\partial x} + \frac{1}{2}\alpha^2 x^2 \frac{\partial^2}{\partial x^2}$ yields:

$$\hat{\mathcal{A}}g(s, x) = -\rho e^{-\rho s}(x - 1) + rxe^{-\rho s} + 0 = e^{-\rho s} [x(r - \rho) + \rho]$$

Setting $\hat{\mathcal{A}}g > 0$, I find:

$$x(\rho - r) < \rho \quad \implies \quad x < \frac{\rho}{\rho - r}$$

Furthermore, for $x \leq 1$, the immediate payoff is 0, but waiting yields a strictly positive expected reward due to the volatility of X_t . Thus, the entire interval $(0, \frac{\rho}{\rho-r})$ has positive expected drift. Because the optimal strategy never stops when the expected drift is positive, this entire interval must be contained within D .

Since D must be connected and anchored at 0, it must take the form $D = \{(s, x) : 0 < x < x_0\}$ for some upper boundary $x_0 > \frac{\rho}{\rho-r}$.

b) Solving the Boundary Value Problem

Inside the continuation region D , the value function $f(s, x)$ must be a martingale, meaning it solves the Dirichlet boundary value problem $\hat{\mathcal{A}}f = 0$:

$$\frac{\partial f}{\partial s} + rx \frac{\partial f}{\partial x} + \frac{1}{2}\alpha^2 x^2 \frac{\partial^2 f}{\partial x^2} = 0 \quad \text{for } 0 < x < x_0$$

with boundary conditions $f(s, 0) = 0$ and $f(s, x_0) = e^{-\rho s}(x_0 - 1)^+$.

I guess a separable solution of the form $f(s, x) = e^{-\rho s} \phi(x)$. Substituting this into the PDE reduces it to a 1-dimensional Cauchy-Euler ODE:

$$-\rho \phi(x) + rx\phi'(x) + \frac{1}{2}\alpha^2 x^2 \phi''(x) = 0$$

I solve this by guessing a polynomial solution $\phi(x) = x^\gamma$. Substituting the derivatives yields the characteristic quadratic equation:

$$\frac{1}{2}\alpha^2 \gamma(\gamma - 1) + r\gamma - \rho = 0 \quad \implies \quad \frac{1}{2}\alpha^2 \gamma^2 + \left(r - \frac{1}{2}\alpha^2\right)\gamma - \rho = 0$$

Because $\rho > 0$, this parabola opens upward and has one positive root ($\gamma_1 > 0$) and one negative root ($\gamma_2 < 0$). The general solution is:

$$\phi(x) = C_1 x^{\gamma_1} + C_2 x^{\gamma_2}$$

Applying the boundary condition at $x \rightarrow 0$, the term x^{γ_2} blows up to infinity. To keep the value function bounded, I must set $C_2 = 0$. Applying the upper boundary condition at x_0 :

$$\phi(x_0) = C_1 x_0^{\gamma_1} = x_0 - 1 \quad \implies \quad C_1 = (x_0 - 1)x_0^{-\gamma_1}$$

Therefore, the bounded solution to the Dirichlet problem is:

$$f(s, x) = e^{-\rho s} (x_0 - 1) \left(\frac{x}{x_0}\right)^{\gamma_1}$$

c) Determining x_0 via the High Contact Principle

The high contact (smooth pasting) principle dictates that to maximize the value function, the spatial derivative of the value function must match the spatial derivative of the payoff function precisely at the boundary x_0 :

$$\left. \frac{\partial f}{\partial x} \right|_{x=x_0} = \left. \frac{\partial g}{\partial x} \right|_{x=x_0}$$

1. **Derivative of the payoff g :** For $x > 1$, $g(s, x) = e^{-\rho s}(x - 1)$, so $\frac{\partial g}{\partial x} = e^{-\rho s}$. 2. **Derivative of the value function f :**

$$\frac{\partial f}{\partial x} = e^{-\rho s} C_1 \gamma_1 x^{\gamma_1 - 1} = e^{-\rho s} (x_0 - 1) x_0^{-\gamma_1} \gamma_1 x^{\gamma_1 - 1}$$

Evaluating exactly at $x = x_0$:

$$\left. \frac{\partial f}{\partial x} \right|_{x=x_0} = e^{-\rho s} \gamma_1 \frac{x_0 - 1}{x_0}$$

Equating the two derivatives:

$$e^{-\rho s} \gamma_1 \frac{x_0 - 1}{x_0} = e^{-\rho s}$$

$$\gamma_1(x_0 - 1) = x_0 \quad \implies \quad x_0(\gamma_1 - 1) = \gamma_1 \quad \implies \quad x_0 = \frac{\gamma_1}{\gamma_1 - 1}$$

Exercise 13

Problem Statement: We seek the optimal stopping time τ to maximize the expected discounted profit from extracting a resource. The price P_t and remaining quantity Q_t follow:

$$dP_t = \alpha P_t dt + \beta P_t dB_t, \quad P_0 = p$$

$$dQ_t = -\lambda Q_t dt, \quad Q_0 = q$$

The running profit rate is $\lambda P_t Q_t - K$, and the terminal bequest function is $g(P_\tau, Q_\tau)$. The optimal value function is:

$$G^*(s, p, q) = \sup_{\tau} \mathbb{E}^{(s, p, q)} \left[\int_0^\tau (\lambda P_t Q_t - K) e^{-\rho(s+t)} dt + e^{-\rho(s+\tau)} g(P_\tau, Q_\tau) \right]$$

Solution:

a) The Characteristic Operator and Variational Inequalities

I augment the state space to handle the time-inhomogeneity and define the diffusion process $X_t = (s+t, P_t, Q_t)^T$. By applying Itô's formula, the characteristic operator $\mathcal{A}$ of X_t applied to a sufficiently smooth function $\Phi(s, p, q)$ is:

$$\mathcal{A} = \frac{\partial}{\partial s} + \alpha p \frac{\partial}{\partial p} - \lambda q \frac{\partial}{\partial q} + \frac{1}{2} \beta^2 p^2 \frac{\partial^2}{\partial p^2}$$

Let $f(s, p, q) = e^{-\rho s}(\lambda p q - K)$ be my running cost, and $G(s, p, q) = e^{-\rho s} g(p, q)$ be my immediate payoff. The variational inequalities require me to find a function G^* and a continuation region D such that: 1. $G^*(s, p, q) \geq G(s, p, q)$ everywhere. 2. $\mathcal{A}G^* + f \leq 0$ everywhere. 3. $\mathcal{A}G^* + f = 0$ strictly inside the continuation region D . 4. Smooth pasting holds at the boundary ∂D .

b) Determining the Positive Drift Region U

Assume the bequest function is exactly the current market value of the remaining resources: $g(p, q) = pq$. Therefore, my immediate stopping payoff is $G(s, p, q) = e^{-\rho s} pq$.

I find the region U where the expected drift of stopping is strictly positive: $U = \{(s, p, q) : \mathcal{A}G + f > 0\}$. I compute $\mathcal{A}G$:

$$\mathcal{A}(e^{-\rho s} pq) = -\rho e^{-\rho s} pq + \alpha p(e^{-\rho s} q) - \lambda q(e^{-\rho s} p) + 0$$

$$\mathcal{A}G(s, p, q) = e^{-\rho s} pq(\alpha - \rho - \lambda)$$

Now, add the running reward $f(s, p, q) = e^{-\rho s}(\lambda p q - K)$:

$$\mathcal{A}G + f = e^{-\rho s} [pq(\alpha - \rho - \lambda) + \lambda p q - K] = e^{-\rho s} [pq(\alpha - \rho) - K]$$

Conclusion (i): If $\rho \geq \alpha$

If the discount rate outpaces or equals the price growth ($\rho \geq \alpha$), then $(\alpha - \rho) \leq 0$. Since price $p \geq 0$, quantity $q \geq 0$, and the fixed cost $K > 0$, the expression $pq(\alpha - \rho) - K$ is strictly negative everywhere. The region U is empty ($U = \emptyset$).

Every second I wait, I strictly lose expected value. Therefore, I must stop immediately:

$$\tau^* = 0 \quad \text{and} \quad G^*(s, p, q) = pqe^{-\rho s}$$

Conclusion (ii): If $\rho < \alpha$

If the price growth outpaces the discount rate, then $(\alpha - \rho) > 0$. The drift is strictly positive when:

$$pq(\alpha - \rho) - K > 0 \quad \implies \quad pq > \frac{K}{\alpha - \rho}$$

Since it is never optimal to stop when the drift is strictly positive, the continuation region D must fully encapsulate this region:

$$D \supset \left\{ (s, p, q); pq > \frac{K}{\alpha - \rho} \right\}$$

c) Solving for G^* when $\rho < \alpha < \rho + \lambda$

Guided by part (b), the decision to stop depends entirely on the total resource value $y = pq$. I guess the continuation region is of the form $D = \{pq > y_0\}$ for some lower threshold y_0 .

Inside D , the value function $\phi(s, p, q) = e^{-\rho s}\psi(y)$ must solve the Dirichlet boundary value problem $\mathcal{A}\phi + f = 0$. Applying the operator to my guess:

$$\mathcal{A}\phi = e^{-\rho s} \left[-\rho\psi(y) + \alpha pq\psi'(y) - \lambda pq\psi'(y) + \frac{1}{2}\beta^2 p^2 q^2 \psi''(y) \right]$$

Substitute $y = pq$ and set $\mathcal{A}\phi + f = 0$:

$$e^{-\rho s} \left[\frac{1}{2}\beta^2 y^2 \psi''(y) + (\alpha - \lambda)y\psi'(y) - \rho\psi(y) + \lambda y - K \right] = 0$$

This gives a non-homogeneous Cauchy-Euler ODE for $\psi(y)$:

$$\frac{1}{2}\beta^2 y^2 \psi''(y) + (\alpha - \lambda)y\psi'(y) - \rho\psi(y) = K - \lambda y$$

Step 1: Particular Solution

I guess a linear form $\psi_p(y) = Ay + B$.

$$(\alpha - \lambda)(Ay) - \rho(Ay + B) = K - \lambda y$$

Matching coefficients yields:

$$A(\alpha - \lambda - \rho) = -\lambda \quad \implies \quad A = \frac{\lambda}{\rho + \lambda - \alpha}$$

$$-\rho B = K \quad \implies \quad B = -\frac{K}{\rho}$$

Step 2: Homogeneous Solution

I guess $\psi_h(y) = y^\gamma$, leading to the characteristic equation:

$$\frac{1}{2}\beta^2\gamma(\gamma-1) + (\alpha-\lambda)\gamma - \rho = 0$$

This parabola opens upward with roots $\gamma_1 > 0$ and $\gamma_2 < 0$. The general solution is $\psi(y) = C_1y^{\gamma_1} + C_2y^{\gamma_2} + \psi_p(y)$. As $y \rightarrow \infty$, the expected value must converge to the particular solution. Thus, $C_1 = 0$.

$$\psi(y) = C_2y^{\gamma_2} + \frac{\lambda}{\rho + \lambda - \alpha}y - \frac{K}{\rho}$$

Step 3: High Contact Principle

At the threshold y_0 , I apply Value Matching ($\psi(y_0) = y_0$) and Smooth Pasting ($\psi'(y_0) = 1$): 1. $C_2y_0^{\gamma_2} + \frac{\lambda}{\rho + \lambda - \alpha}y_0 - \frac{K}{\rho} = y_0$ 2. $C_2\gamma_2y_0^{\gamma_2-1} + \frac{\lambda}{\rho + \lambda - \alpha} = 1$

From (2), isolate $C_2y_0^{\gamma_2}$:

$$C_2y_0^{\gamma_2} = y_0 \left(1 - \frac{\lambda}{\rho + \lambda - \alpha} \right) \frac{1}{\gamma_2} = y_0 \left(\frac{\rho - \alpha}{\rho + \lambda - \alpha} \right) \frac{1}{\gamma_2}$$

Substitute this into (1) and solve for y_0 :

$$y_0 = \frac{K}{\rho} \left(\frac{\rho + \lambda - \alpha}{\rho - \alpha} \right) \left(\frac{\gamma_2}{1 - \gamma_2} \right)$$

Thus, $G^* = e^{-\rho s}\psi(pq)$ inside D , and the optimal stopping time is $\tau^* = \inf\{t > 0 : P_t Q_t \leq y_0\}$.

Exercise 14

Problem Statement: Solve the optimal stopping problem to invest in a project, maximizing the expected discounted net profit:

$$G^*(s, p) = \sup_{\tau} \mathbb{E}^{(s, p)} \left[\int_{\tau}^{\infty} e^{-\rho(s+t)} P_t dt - C e^{-\rho(s+\tau)} \right]$$

where the profit rate P_t follows geometric Brownian motion:

$$dP_t = \alpha P_t dt + \beta P_t dB_t; \quad P_0 = p$$

The constants satisfy $0 < \alpha < \rho$ and $C > 0$.

Solution:

a) Problem Transformation

I decompose the integral from τ to ∞ into two parts:

$$\int_{\tau}^{\infty} e^{-\rho(s+t)} P_t dt = e^{-\rho s} \left[\int_0^{\infty} e^{-\rho t} P_t dt - \int_0^{\tau} e^{-\rho t} P_t dt \right]$$

Because P_t is a geometric Brownian motion, its expected value at time t is $\mathbb{E}^{(s,p)}[P_t] = pe^{\alpha t}$. I can evaluate the expectation of the infinite integral directly:

$$\mathbb{E}^{(s,p)} \left[\int_0^\infty e^{-\rho t} P_t dt \right] = \int_0^\infty e^{-\rho t} (pe^{\alpha t}) dt = p \int_0^\infty e^{-(\rho-\alpha)t} dt = \frac{p}{\rho-\alpha}$$

Substituting this back into the original supremum, I can pull the deterministic expectation outside the supremum operator:

$$G^*(s, p) = e^{-\rho s} \frac{p}{\rho-\alpha} + \Phi(s, p)$$

where $\Phi(s, p)$ is the standard optimal stopping problem:

$$\Phi(s, p) = \sup_{\tau} \mathbb{E}^{(s,p)} \left[\int_0^\tau (-P_t) e^{-\rho(s+t)} dt - C e^{-\rho(s+\tau)} \right]$$

b) The Characteristic Operator and the Continuation Region D

I define the time-space process $\mathbf{Y}_t = (s+t, P_t)$. My running cost is $f(s, p) = -pe^{-\rho s}$ and my terminal payoff is $g(s, p) = -C e^{-\rho s}$.

The characteristic operator $\mathcal{A}$ for $\mathbf{Y}_t$ is:

$$\mathcal{A} = \frac{\partial}{\partial s} + \alpha p \frac{\partial}{\partial p} + \frac{1}{2} \beta^2 p^2 \frac{\partial^2}{\partial p^2}$$

I first locate the region U where the expected drift of the payoff is strictly positive: $U = \{(s, p) : \mathcal{A}g + f > 0\}$. Applying the operator to the terminal payoff g and adding f :

$$\mathcal{A}g(s, p) + f(s, p) = e^{-\rho s} (\rho C - p)$$

Setting this strictly greater than zero yields:

$$\rho C - p > 0 \quad \implies \quad p < \rho C$$

The continuation region D must take the form $D = \{(s, p) : p < p_0\}$ for some threshold $p_0 > \rho C$.

c) Solving the Dirichlet Boundary Value Problem

Inside D ($0 < p < p_0$), the candidate value function $\phi(s, p)$ must be a martingale, meaning it solves $\mathcal{A}\phi + f = 0$. I guess a separable solution $\phi(s, p) = e^{-\rho s} \psi(p)$ and substitute it into the PDE:

$$e^{-\rho s} \left[-\rho \psi(p) + \alpha p \psi'(p) + \frac{1}{2} \beta^2 p^2 \psi''(p) - p \right] = 0$$

This yields the non-homogeneous Cauchy-Euler ODE:

$$\frac{1}{2} \beta^2 p^2 \psi''(p) + \alpha p \psi'(p) - \rho \psi(p) = p$$

1. Particular Solution: I guess $\psi_p(p) = kp$. Substituting this into the ODE gives $k = \frac{-1}{\rho - \alpha}$. So, $\psi_p(p) = -\frac{p}{\rho - \alpha}$.

2. Homogeneous Solution: I guess $\psi_h(p) = p^\gamma$, leading to the characteristic equation:

$$\frac{1}{2}\beta^2\gamma(\gamma - 1) + \alpha\gamma - \rho = 0$$

This upward-opening parabola has one positive root ($\gamma_1 > 1$) and one negative root ($\gamma_2 < 0$). To maintain boundedness at $p \rightarrow 0$, I must set the coefficient for the negative root to 0. Thus:

$$\psi(p) = Ap^{\gamma_1} - \frac{p}{\rho - \alpha}$$

d) High Contact Principle and the Optimal Threshold

At the optimal investment boundary p_0 , I apply Value Matching and Smooth Pasting between ϕ and the payoff g : 1. **Value Matching:** $\psi(p_0) = -C$

$$Ap_0^{\gamma_1} - \frac{p_0}{\rho - \alpha} = -C$$

2. **Smooth Pasting:** $\psi'(p_0) = 0$

$$A\gamma_1 p_0^{\gamma_1 - 1} - \frac{1}{\rho - \alpha} = 0 \quad \implies \quad Ap_0^{\gamma_1} = \frac{p_0}{\gamma_1(\rho - \alpha)}$$

Substitute the smooth pasting result into the value matching equation to eliminate A and solve for p_0 :

$$p_0 = C \frac{\gamma_1(\rho - \alpha)}{\gamma_1 - 1}$$

The optimal investment time is $\tau^* = \inf\{t > 0 : P_t \geq p_0\}$, and my maximal expected discounted net profit is:

$$G^*(s, p) = \begin{cases} Ae^{-\rho s} p^{\gamma_1} & \text{for } p < p_0 \\ e^{-\rho s} \left(\frac{p}{\rho - \alpha} - C \right) & \text{for } p \geq p_0 \end{cases}$$

Exercise 15

Problem Statement: Let B_t be a 1-dimensional Brownian motion and $\rho > 0$ a constant. a) Show that the family $\{e^{-\rho\tau} B_\tau; \tau \text{ stopping time}\}$ is uniformly integrable w.r.t. P^x . b) Solve the optimal stopping problem

$$g^*(s, x) = \sup_{\tau} \mathbb{E}^{(s, x)} \left[e^{-\rho(s+\tau)} (B_\tau - a) \right]$$

when $a > 0$ is constant.

Solution:

a) Proof of Uniform Integrability

To prove that a family of random variables is uniformly integrable, a sufficient condition is to show that the entire family is bounded by a single integrable random variable.

Let Y be the absolute supremum of the discounted process over all time:

$$Y = \sup_{t \geq 0} e^{-\rho t} |B_t|$$

If I can prove that $\mathbb{E}^x[Y] < \infty$, then for *any* stopping time τ , I have $|e^{-\rho\tau} B_\tau| \leq Y$. Because the entire family is dominated by an integrable random variable Y , the family is uniformly integrable.

I evaluate the expectation of Y starting from $B_0 = x$:

$$\mathbb{E}^x[Y] = \mathbb{E}^0 \left[\sup_{t \geq 0} e^{-\rho t} |x + B_t| \right] \leq |x| + \mathbb{E}^0 \left[\sup_{t \geq 0} e^{-\rho t} |B_t| \right]$$

To bound the supremum, I discretize time into integer intervals $[n, n+1]$:

$$\mathbb{E}^0 \left[\sup_{t \geq 0} e^{-\rho t} |B_t| \right] \leq \sum_{n=0}^{\infty} e^{-\rho n} \mathbb{E}^0 \left[\sup_{0 \leq t \leq n+1} |B_t| \right]$$

Using the standard L^2 maximal inequality for Brownian motion, and Jensen's inequality:

$$\mathbb{E}^0 \left[\sup_{0 \leq t \leq n+1} |B_t| \right] \leq 2\sqrt{n+1}$$

Substitute this back into the sum:

$$\mathbb{E}^x[Y] \leq |x| + \sum_{n=0}^{\infty} e^{-\rho n} 2\sqrt{n+1}$$

Because $\rho > 0$, the exponential decay heavily dominates the polynomial growth. By the ratio test, this infinite series converges to a finite constant. Thus, $\mathbb{E}^x[Y] < \infty$, proving that the family is uniformly integrable.

b) Solving the Optimal Stopping Problem

I define the time-space process $\mathbf{Y}_t = (s+t, B_t)$ starting at (s, x) . My immediate payoff function is $g(s, x) = e^{-\rho s}(x - a)$.

1. The Characteristic Operator and the Continuation Region D

The characteristic operator $\mathcal{A}$ for $\mathbf{Y}_t$ is:

$$\mathcal{A} = \frac{\partial}{\partial s} + \frac{1}{2} \frac{\partial^2}{\partial x^2}$$

I evaluate the region U where the expected drift of the payoff is strictly positive:

$$\mathcal{A}g(s, x) = -\rho e^{-\rho s}(x - a)$$

Setting this strictly greater than zero yields $x < a$. I guess the continuation region takes the form $D = \{(s, x) : x < x_0\}$ for some boundary $x_0 \geq a$.

2. Solving the Dirichlet Boundary Value Problem

Inside the continuation region D , I guess a separable solution $\phi(s, x) = e^{-\rho s}\psi(x)$ and substitute it into $\mathcal{A}\phi = 0$:

$$\frac{1}{2}\psi''(x) - \rho\psi(x) = 0$$

This is a linear homogeneous ODE with roots $\gamma = \pm\sqrt{2\rho}$. The general solution is:

$$\psi(x) = C_1 e^{x\sqrt{2\rho}} + C_2 e^{-x\sqrt{2\rho}}$$

As $x \rightarrow -\infty$, to keep the function bounded from below, I must set $C_2 = 0$.

$$\psi(x) = C_1 e^{x\sqrt{2\rho}}$$

3. The High Contact Principle and the Optimal Boundary

At the optimal threshold x_0 , I apply Value Matching ($\psi(x_0) = x_0 - a$) and Smooth Pasting ($\psi'(x_0) = 1$): 1. $C_1 e^{x_0\sqrt{2\rho}} = x_0 - a$ 2. $C_1 \sqrt{2\rho} e^{x_0\sqrt{2\rho}} = 1$

To find x_0 , I divide the Smooth Pasting equation by the Value Matching equation and solve:

$$x_0 = a + \frac{1}{\sqrt{2\rho}}$$

The optimal stopping time is $\tau^* = \inf\{t > 0 : B_t \geq x_0\}$, and the globally optimal value function is:

$$g^*(s, x) = \begin{cases} e^{-\rho s} \frac{1}{\sqrt{2\rho}} e^{(x-x_0)\sqrt{2\rho}} & \text{for } x < a + \frac{1}{\sqrt{2\rho}} \\ e^{-\rho s} (x - a) & \text{for } x \geq a + \frac{1}{\sqrt{2\rho}} \end{cases}$$

Exercise 16

Problem Statement: Solve the optimal stopping problem

$$G^*(s, p) = \sup_{\tau} \mathbb{E}^{(s, p)} \left[\int_{\tau}^{\infty} e^{-\rho(s+t)} P_t dt - C e^{-\rho(s+\tau)} \right]$$

where $dP_t = \mu dt + \sigma dB_t$; $P_0 = p$.

Solution:

a) Problem Transformation

I split the infinite integral into two components:

$$\int_{\tau}^{\infty} e^{-\rho(s+t)} P_t dt = e^{-\rho s} \left[\int_0^{\infty} e^{-\rho t} P_t dt - \int_0^{\tau} e^{-\rho t} P_t dt \right]$$

Because P_t is an arithmetic Brownian motion, its expected value at time t is $\mathbb{E}^{(s, p)}[P_t] = p + \mu t$. I evaluate the expectation of the infinite integral directly:

$$\mathbb{E}^{(s, p)} \left[\int_0^{\infty} e^{-\rho t} P_t dt \right] = \frac{p}{\rho} + \frac{\mu}{\rho^2}$$

Substituting this back into my supremum, I extract it to define a standard optimal stopping problem $\Phi(s, p)$:

$$G^*(s, p) = e^{-\rho s} \left(\frac{p}{\rho} + \frac{\mu}{\rho^2} \right) + \Phi(s, p)$$

where:

$$\Phi(s, p) = \sup_{\tau} \mathbb{E}^{(s, p)} \left[\int_0^{\tau} (-P_t) e^{-\rho(s+t)} dt - C e^{-\rho(s+\tau)} \right]$$

b) The Characteristic Operator and the Continuation Region D

I define the time-space process $\mathbf{Y}_t = (s + t, P_t)$. My running cost is $f(s, p) = -p e^{-\rho s}$ and my terminal payoff is $g(s, p) = -C e^{-\rho s}$.

The characteristic operator $\mathcal{A}$ for the arithmetic process $\mathbf{Y}_t$ is:

$$\mathcal{A} = \frac{\partial}{\partial s} + \mu \frac{\partial}{\partial p} + \frac{1}{2} \sigma^2 \frac{\partial^2}{\partial p^2}$$

I locate the region U where the expected drift of stopping immediately is strictly positive ($\mathcal{A}g + f > 0$):

$$\mathcal{A}g(s, p) + f(s, p) = e^{-\rho s}(\rho C - p) > 0 \implies p < \rho C$$

The continuation region D must take the form $D = \{(s, p) : p < p_0\}$ for some optimal threshold $p_0 > \rho C$.

c) Solving the Dirichlet Boundary Value Problem

Inside D ($p < p_0$), the candidate value function $\phi(s, p)$ solves $\mathcal{A}\phi + f = 0$. I guess a separable solution $\phi(s, p) = e^{-\rho s} \psi(p)$ and substitute it into the PDE:

$$\frac{1}{2} \sigma^2 \psi''(p) + \mu \psi'(p) - \rho \psi(p) = p$$

1. Particular Solution: I guess a linear form $\psi_p(p) = Ap + B$. Matching coefficients yields $A = -\frac{1}{\rho}$ and $B = -\frac{\mu}{\rho^2}$. So, $\psi_p(p) = -\frac{p}{\rho} - \frac{\mu}{\rho^2}$.

2. Homogeneous Solution: I guess $\psi_h(p) = e^{\gamma p}$, leading to the characteristic equation:

$$\frac{1}{2} \sigma^2 \gamma^2 + \mu \gamma - \rho = 0$$

This upward-opening parabola has one positive root ($\gamma_1 > 0$) and one negative root ($\gamma_2 < 0$). To maintain mathematical bounded behavior as $p \rightarrow -\infty$, I must set the coefficient for γ_2 to 0.

$$\psi(p) = C_1 e^{\gamma_1 p} - \frac{p}{\rho} - \frac{\mu}{\rho^2}$$

d) High Contact Principle and the Optimal Threshold

At the boundary p_0 , I apply Value Matching ($\psi(p_0) = -C$) and Smooth Pasting ($\psi'(p_0) = 0$): 1. $C_1 e^{\gamma_1 p_0} - \frac{p_0}{\rho} - \frac{\mu}{\rho^2} = -C$ 2. $C_1 \gamma_1 e^{\gamma_1 p_0} - \frac{1}{\rho} = 0 \implies C_1 e^{\gamma_1 p_0} = \frac{1}{\rho \gamma_1}$

Substitute the smooth pasting result into the value matching equation to eliminate C_1 :

$$\frac{1}{\rho\gamma_1} - \frac{p_0}{\rho} - \frac{\mu}{\rho^2} = -C$$

Multiply the entire equation by ρ and isolate p_0 :

$$p_0 = \rho C + \frac{1}{\gamma_1} - \frac{\mu}{\rho}$$

e) Finalizing the Optimal Value Function G^*

I reconstruct my original objective function: $G^*(s, p) = e^{-\rho s} \left(\frac{p}{\rho} + \frac{\mu}{\rho^2} \right) + e^{-\rho s} \psi(p)$ for $p < p_0$.

The optimal investment time is $\tau^* = \inf\{t > 0 : P_t \geq p_0\}$, and the maximal expected discounted net profit is:

$$G^*(s, p) = \begin{cases} C_1 e^{-\rho s} e^{\gamma_1 p} & \text{for } p < p_0 \\ e^{-\rho s} \left(\frac{p}{\rho} + \frac{\mu}{\rho^2} - C \right) & \text{for } p \geq p_0 \end{cases}$$